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An Interdisciplinary Review of the Experimental Evidence on How Humans Interact with Machines

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We and It:
An interdisciplinary review of the experimental evidence on how humans interact with machines *

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Abstract

Today, humans interact with automation frequently and in a variety of settings ranging from private to professional. Their behavior in these interactions has attracted considerable research interest across several fields, with sometimes little exchange among them and seemingly inconsistent findings. In this article, we review 136 experimental studies on how people interact with automated agents, that can assume different roles. We synthesize the evidence, suggest ways to reconcile inconsistencies between studies and disciplines, and discuss organizational and societal implications. The reviewed studies show that people react to automated agents differently than they do to humans: In general, they behave more rationally, and are less prone to emotional and social responses. Task context, performance expectations and the distribution of decision authority between humans and automated agents are all factors that systematically impact the willingness to accept automated agents in decision-making - that is, humans seem willing to (over-)rely on algorithmic support, yet averse to fully ceding their decision authority. The impact of these behavioral regularities for the deliberation of the benefits and risks of automation in organizations and society is discussed.

Keywords: automation, human-computer interaction, human-machine interaction, algorithmic decision making, experimental evidence, literature review.
JEL-classification: O33, C90, D90.

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1 Introduction

Early in the morning, James drives to work on the route picked out for him by his navigation system. On the highway, his car's blind spot assistant helps him avoid a potentially dangerous situation. He happily listens to a playlist of new songs selected for him by an app. An alert on his phone warns him about unusual activity on his bank account, but the issue is swiftly resolved with the help of the bank's chat bot. After passing the automated barrier to enter his office, James plugs his portable computer into the docking station on his desk, and starts his day.

While James's morning may seem like nothing out of the ordinary, it serves to illustrate something rather extraordinary: how frequent and how routine interactions between humans and automated agents (i.e., agents of non-human nature powered by technology) have become. Had someone read this paragraph a few decades ago, they might have assumed that our hero was James Bond. In contrast, today, automated agents are everywhere. They collaborate with humans in the workplace and in some cases even occupy managerial positions. They engage in sales negotiations with humans, and trade with them on the stock market. They are involved in deciding whether a human should receive a job or be released from jail. And, the recent move into a far more digital way of working due to the global pandemic has increased the involvement of automated agents in professional life and decision-making even further.

What happens when interactions between humans turn into interactions between humans and increasingly capable automated agents? How do humans react to the growing involvement of automated agents in decision-making? The ubiquity of automation in today's world calls for an interdisciplinary effort to answer these questions [Rahwan et al., 2019]. At the same time, the fast-growing number of studies, the wide range of methodologies used and a number of (seemingly) inconsistent findings make it difficult for interested researchers or decision-makers to gain an understanding of the existing evidence. In this review, we attempt to change this by synthesizing the findings of 136 experimental studies from a wide range of disciplines into a number of generalizable findings which describe how humans behave when they interact with automated agents. We also point out a number of questions which the existing evidence does not yet allow to answer, because of a lack of data or because of inconsistent results, and provide concrete suggestions for future research.

As this review will show, a number of behavioral regularities emerge from the empirical evidence on human-machine interactions. First, a series of experiments document that humans interact with automated agents in social ways, yet differently than they do with fellow humans: In general, they exhibit less of an emotional response to the actions of automated agents, and are less concerned - but not completely unconcerned - with social rules of conduct when interacting with automated agents. This reduced emotional and social response can improve outcomes in interactions that benefit from increased rationality - for instance, people have been found to be more likely to reveal sensitive health information to an automated system. On the other hand, this effect can be harmful where emotions or social rules are crucial drivers and motivations for beneficial outcomes - for example, interacting with automated agents has been shown to increase the likelihood of engaging in unethical or self-serving behavior. Certain design features of automated agents may affect the level of emotional and social response they trigger: For instance, some experiments will show that features which cause automated agents to be perceived as more human-like intensify the emotional and social response to these agents. Task type and context seem to matter: While experiments document that humans are willing to engage with automated agents in contexts perceived as analytical or objective, they seem reluctant to do so in more social or moral contexts. In addition to these general results, we discuss insights from a number of studies where automated agents assume specific roles in interactions of particular relevance for the workplace.

We structure the review by the type of role the automated agents may play: first, as relatively independent agents who either collaborate with humans as their peers, or manage them as their superiors. Though this use of automated agents is growing, experimental studies on this type of interaction remain rare. For example, the existing evidence establishes that having an automated teammate crucially affects worker (and hence team) performance, yet documents multiple channels for this effect which may lead to both an increase and decrease of performance. With regards to automated agents taking managerial roles, we discuss studies which show that humans seem willing to accept automated managers - and, indeed, that automated managers may even have an advantage over humans in contexts perceived as analytical. In addition, we consider interactions where automated-agents are involved in decision-making situations. Here the existing evidence seems inconsistent at first glance: some studies show that humans are averse to delegating decision authority to automated agents, others find them to be appreciative of automated advice, and sometimes even overreliant on automated decision-making supports. We discuss why this could be the case - again, separating the evidence by the specific role the automated agents play, namely that of a delegate or of a decision aid, will help here - and propose a number of concrete testable hypotheses to reconcile these findings. In particular, we will argue that the acceptance of automated decision-making depends on the distribution of decision authority, the context of the decision and on performance expectations.

The studies covered in this review are chosen according to two criteria: first, we cover studies which investigate what changes when humans interact with automated agents, with the benchmark being human-human interactions. Choosing to focus on the presence of automation in an interaction, we use a broad definition of what constitutes an "automated agent", without imposing any constraints either on the underlying technology nor on its sophistication (indeed, computers/computer systems, machines, robots, algorithms, AI systems, intelligent agents and so on can all constitute automated agents). This allows us to draw on a large range of studies spanning from the early years of automation up until recent discussions of sophisticated machine learning algorithms. It goes without saying that the increased sophistication of automation and the growing familiarity of people with technology over the last decades may affect the way in which people interact with it. Interestingly however, many of the behaviors reported in this paper appear to be triggered by the switch from purely human-human interactions to human-automation interactions, rather than by specific aspects of the underlying technology. Our second selection criterion is that we cover studies which use experimental methods to investigate human behavior in human-automated agent interactions, with the aim to discuss causal effects as well as underlying mechanisms. This focus on human behavior *in interactions* implies that we do not discuss research e.g. on algorithm design, user experience (UX) or interface design, nor on the stated attitudes towards automation in general, the macroeconomic consequences of automation or on the ethics of automation. We do touch upon some heavily debated topics such as explainable AI or algorithmic fairness in the discussion for additional perspectives on the topic.

Following our goal of identifying underlying behavioral regularities regardless of disciplinary borders, our review covers a wide range of fields, including economics, management, psychology, marketing and consumer research, sociology, judgement and decision making, human-computer interaction, neuroscience, computer science, information systems, medicine and even aeronautics. As a result of the differences across these fields, our sample includes studies which elicit stated preferences, revealed preferences or use physiological measurements, which use or do not use incentives, and which resort to various subject pools. The Appendix provides short notes on the methodologies of all cited experiments, observational studies and literature reviews.

To sum up our selection strategy, when faced with the trade-off between breadth or depth, we opted for the former. Regarding the methodology used to identify the relevant studies, due to the important differences in terminology and keywords across disciplines and technologies, our above-described choices implied that the search of digital databases using keywords soon proved to be unsuitable. Instead, we chose to primarily resort to ancestry searching (alternatively called backward reference searching). That is, after we identified a particular study as relevant, we considered if the studies it cites may be relevant for the review, too. If several relevant studies appeared in the same outlet, we more systematically reviewed other publications in that outlet resorting to journal hand searching. Finally, we consulted with other researchers working in the field [following Whittemore and Knafl, 2005]. To guard against publication bias and to include the most recent findings we deliberately include publicly available working papers [following Thompson and Pocock, 1991]. It is important to note that, given the ambitious goal of the review and the number of covered disciplines, the list of included papers is by no means exhaustive, but covers main themes and will hopefully provide a useful starting point to an interested reader.

The contribution of this review is threefold. First, we provide an extensive overview of behavioral research spanning from early automation research to discussions of modern AI systems, in a form that may prove useful to both interested scientists and decision-makers in the field. We synthesize the evidence from numerous studies into a number of recurrent findings. Second, we scrutinize a number of seemingly inconsistent findings, and suggest a structured way to reconcile them. We are able to address such inconsistencies due to the broad range of experiments covered (for other literature reviews that focus in more detail on only specific strands of literature see e.g. Jussupow et al. [2020] on algorithm aversion and appreciation, Burton et al. [2020] on algorithm aversion, Parasuraman and Manzey [2010], Alberdi et al. [2009] on automation bias and automation-induced complacency or March [2019] on the use of computer players in economic experiments). With this, we hope to provide concrete and testable hypotheses which might allow future research to investigate the currently open question of why different experiments document very different reactions to automated agents. Finally, we link the empirical findings to the ongoing societal discussion about the benefits and harms of automation and discuss how behavioral research can inform effective regulation of automated decision-making and introduction of automation in the workplace. With all of this, we aim to promote the dissemination of insights and ideas across disciplines, and to help provide grounds for evidence-based decision-making.

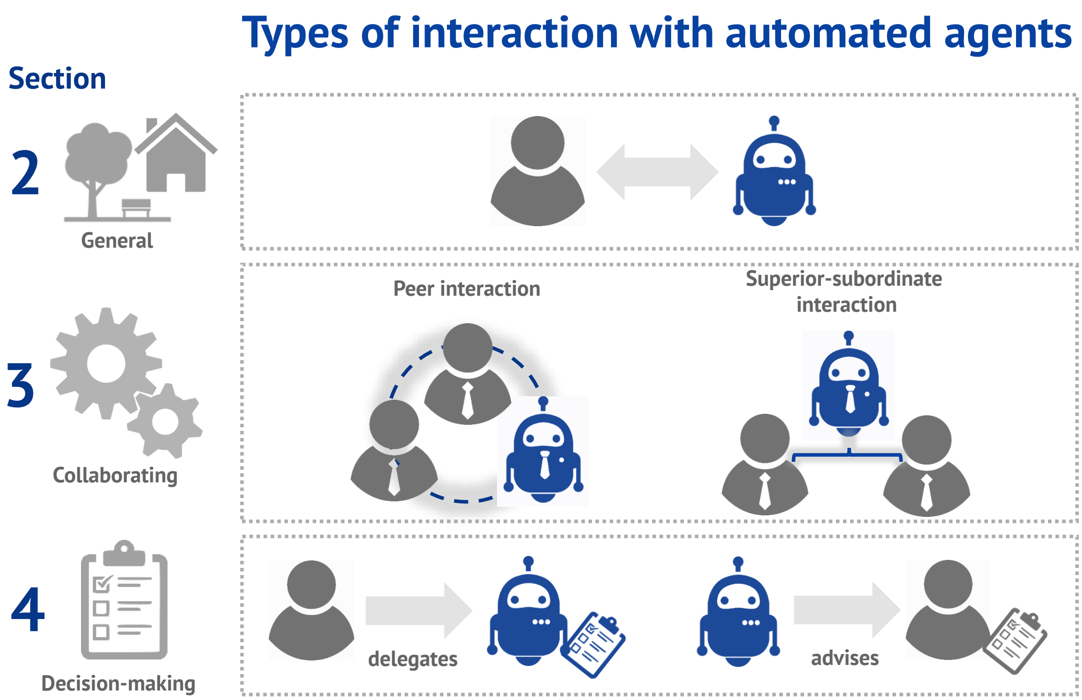

Figure 1: TYPES OF INTERACTION BETWEEN HUMANS AND AUTOMATED AGENTS Illustration of different categories of interactions and the sections of the review which covers them. Within the collaboration section, interactions are separated according to the role of the automated agent as a peer (middle left) or superior (middle right). Within the decision-making section, the interactions are separated according to the distribution of agency within the interaction, with the human delegating the decision authority to the automated agent (bottom left), or the automated agent advising the human who retains the authority over the decision (bottom right).

Fig.1 provides an overview of the structure of the paper and shows the kinds of interactions considered in the review. Section 2 starts by discussing experiments investigating general patterns in interactions between humans and automated agents. In Sections 3 and 4, we consider specific types of interactions: Section 3 focuses on interactions in which automated agents collaborate with humans in the workplace, either as their peers and teammates or as their superiors. Section 4 discusses experiments which investigate how humans react to the involvement of automated agents in decision-making, where automated agents act either as delegates or as decision aids. At the end of each subsection, we offer short summaries of the key findings of the mentioned studies. Section 5 draws upon findings of the review to offer some thoughts on future avenues of research by linking the existing evidence to the ongoing political and societal discussion about the risks and benefits of automation. Finally, section 6 concludes.

2 Social interactions with automated agents

2.1 The perception of automated agents as social interaction partners

To start our discussion, let us first establish whether interactions between humans and automated agents can appropriately be considered as social interactions. Researchers have long documented that humans tend to create narratives around events [Bruner, 1991], and for instance attribute agency to inanimate objects [Heider and Simmel, 1944]. In an early series of lab experiments in the aptly termed *computers are social actors (CASA)* paradigm, psychologists documented that people apply social rules and expectations to computers [for an overview see Nass and Moon, 2000, Reeves and Nass, 1996]. Participants in these experiments applied gender stereotypes [Nass et al., 1997] and ethnic stereotypes [Nass et al., 1996] to computers. They reacted to automated feedback [De Laere et al., 1998], preferred computers arbitrarily marked as teammates [Nass et al., 1996] and reciprocated helpful acts by computers [Fogg and Nass, 1997]. Reactions of reciprocity were even adjusted to prevailing cultural norms [Katagiri et al., 2001]. Yet, when asked directly, most participants in these experiments denied that they felt that computers had a personality, nor that they felt computers warranted social or polite treatment [Nass and Moon, 2000].

Later experiments provided further evidence for the social treatment of machines. Participants were again found to apply gender stereotypes [Tay et al., 2013, Eyssel and Hegel, 2012, Tay et al., 2014] and racial stereotypes [Bartneck et al., 2018] to automated agents. They were also found to use social cues like smiles and silence fillers when interacting with automated agents [Aharoni and Fridlund, 2007], and to react to automated flattery [Lee, 2010]. Given sufficient behavioral realism, social reactions can be triggered by both avatars (digital representations of humans) and virtual agents (interactive computer programs) [Von der Puetten et al., 2010]. A recent experiment even showed that robots can trigger actions of social conformity at a similar rate as humans - but only until participants lose trust in faulty robots [Salomons et al., 2018].

Importantly, these findings do not imply that social interactions between humans and automated agents are equal to those between humans - as this review will discuss at length. Indeed, the difference in how humans interact with fellow humans or with automated agents are visible in neurophysiological studies, which document that different areas of the brain are activated in these different interactions [see e.g. Krach et al., 2008, Chaminade et al., 2012, Sanfey et al., 2003, McCabe et al., 2001]. The reason for this might lie in the way humans perceive their respective counterparts' agency. In social interactions people engage in the process of *mentalizing* to infer the mental state and capacities of their counterparts [Frith and Frith, 2006]. This deliberation seems to occur less with automated agents: the area of the brain connected to mentalizing is activated less when humans interact with automated agents [McCabe et al., 2001, Coricelli and Nagel, 2009]. The reduced need to infer the mental state of one's counterpart might also explain the repeated finding that people react faster to actions of automated counterparts [see e.g. Coricelli and Nagel, 2009, Farjam and Kirchkamp, 2018].

And, indeed, as a large survey shows, people do not attribute robots with minds [Gray et al., 2007]. This survey in particular showed that humans deny robots the capacity to experience moral right or wrong [Gray et al., 2007]. People even report feelings of unease when robots are described as able to feel [Gray and Wegner, 2012]. However, despite this perception, people seem averse to mistreating automated agents. Indeed, people were found to exhibit both physiological and behavioral displays of stress when watching videos of robots being ”tortured” [Rosenthal-von der Pütten et al., 2013], or when asked to administer ”painful” electric shocks to robots [Bartneck et al., 2005] or virtual agents [Slater et al., 2006] in two Milgram-style experiments [Milgram, 1963]. Another experiment showed that people were unwilling to follow a command to destroy a tower a robot had built if the robot protested and started ”sobbing” [Briggs and Scheutz, 2014].

Summary of section 2.1 Experimental studies have shown that people interact with automated agents in social ways, but differently than with other humans: When interacting with automated agents, people engage less in mentalizing (i.e. inferring the mental state of the counterpart). People do not seem to attribute robots with 'minds' or feelings, but are nevertheless found to be averse to mistreating automated agents.

2.2 The reduced emotional and social response to automated agents

A robust finding which documents the differences between human-human and human-automated agent interactions is that interacting with automated agents triggers a reduced emotional and social response. This phenomenon has been documented using both subjective and behavioral measures, e.g. in trust games [Schniter et al., 2020, Lim and Reeves, 2010] or in ultimatum, dictator and public goods games [Melo et al., 2016]. It has also been found using physiological measurements, e.g. when playing computer games [Mandryk et al., 2006], in ultimatum games [Sanfey et al., 2003, van't Wout et al., 2006] or in auctions [Teubner et al., 2015, Adam et al., 2015].

The reduced emotional response to automated agents is manifested both in terms of a less immediate emotional reaction to the agent's actions as well as in a generally decreased level of emotional arousal in human-computer interactions [as demonstrated e.g. in Teubner et al., 2015]. Interacting with an automated counterpart seems to narrow the entire emotional spectrum: people react less positively to desirable actions by automated agents, and less negatively to undesirable ones. For instance, principals who delegated tasks reported significantly less enjoyment of good outcomes and significantly less anger for bad outcomes when the task was delegated to an algorithm rather than to a human [Leyer and Schneider, 2019]. In another study, observers rated a physician's correct medical decisions less positively and incorrect decisions less negatively if the physician was described to use an automated decision-making aid [Pezzo and Pezzo, 2006]. In a social exchange experiment, participants perceived coercive actions by computers as less unjust than coercive actions by humans, and retaliated against them less. In contrast, cooperative actions were perceived as similarly just regardless of who made them [Shank, 2012]. The reduced emotional response could be due to a perceived lack of intent on part of the automated agent [as speculated by Shank, 2012]. Hidalgo et al. [2021] also highlight the role of intentions: in an investigation of almost 90 vignette scenarios with moral tasks, the authors find that identical decisions are judged differently depending on whether they are made by a human or an automated agents. The perception of intent seems to drive this difference: people seem more likely to judge humans by their intentions and machines by their outcomes, resulting in more forgiving judgements towards automated agents in scenarios involving fairness, loyalty, and labor displacement.

A potential economic upside of this decreased emotional and social response is the increased rationality of interactions. The introduction of automated agents has, hence, been shown to reduce bubbles in simulated stock markets [Farjam and Kirchkamp, 2018], to increase bargaining efficiency in auctions [Adam et al., 2018] or to improve cooperation in ultimatum games [Erlei et al., 2020]. Computers can also help avoid undesirable social responses, and, for example, increase the willingness to disclose uncomfortable information [Lucas et al., 2014]. Reporting to computers has even been shown to significantly increase the likelihood of disclosures of intimate partner violence [Ahmad et al., 2009, Humphreys et al., 2011]. On the downside, automation can prove harmful in contexts where beneficial behaviors are driven by emotions and social concerns like pro-sociality, social comparisons or reciprocity. An experiment which demonstrates this has participants play public goods, ultimatum and dictator games against other humans or against automated agents [Melo et al., 2016]. In all three games, participants prove less willing to share with automated counterparts, and feel less guilty about exploiting them. This mirrors findings of earlier CASA studies, which documented participants' willingness to engage in self-serving bias with automated counterparts (i.e., to attribute positive outcomes to themselves and negative outcomes to the computer) [Moon and Nass, 1998, Moon, 2003]. The lack of altruism and social pressure towards automated teammates was also shown to lower worker's productivity in an experimental sequential assembly line task [Corgnet et al., 2019]. Further studies document that the presence of automated agents can increase the willingness to engage in unethical or even illegal behavior [for a review see Köbis et al., 2021]. For instance, when given a chance to misreport the outcome of a coin toss to increase their monetary profit, participants were significantly more likely to lie when reporting to a computer [Cohn et al., 2018]. In summary, while automation can have beneficial effects in contexts where emotions or social concerns are detrimental, it can also be harmful by reducing them in contexts where they are beneficial.

Yet, the social and emotional response to automated agents is mediated by, for instance, cultural background of a human counterpart [for overview see Lim et al., 2021] and by certain design features of automated agents that can render them to be perceived more human-like and emotionally intelligent. An important factor which affects a person's emotional and social response to an automated agent seems to be the agent's behavior and appearance. Indeed, humans seem to prefer interacting with automated agents which display contingent verbal and non-verbal reactions [Gratch et al., 2007], relational and animated behaviors [Bickmore et al., 2005, Nitsch and Glassen, 2015] and group emotions [Correia et al., 2018].Unexpected behaviors by robots can also amplify social responses: for example, people were more likely to display social reactions towards robots that unexpectedly cheated in a rock-paper-scissors game [Short et al., 2010]. In a study with service robots who delivered medicine to elderly patients, the use of human-like faces and voices where found to significantly promote positive emotional responses to the robots [Zhang et al., 2010]. People were also found to report feeling more comfortable around human-like robots and to perceive them as more useful [Castelo, 2019]. The human-like appearance of an agent even affected the attribution of credit and blame in human-robot teams: automated teammates who appeared more human-like were relied upon more and attributed more credit for the output than those who didn't [Hinds et al., 2004]. However, further studies show that the effects of human-like appearances remain unclear. An interesting counterpoint comes from an experiment which shows that anthropomorphic robots trigger more compensatory consumer responses - yet, not because people liked the human-like robots, but because they felt discomfort in their presence and perceived them as a threat to human identity [Mende et al., 2019]. And, as was mentioned earlier, participants in another study disliked robots who were described as able to feel [Gray and Wegner, 2012]. More generally, a study on the roots of the behaviors documented by the CASA paradigm failed to find support for the claim that people are more likely to react socially to anthropomorphic characters [Lee, 2010]. Nishio et al. [2018] also find that it is not the appearance of the counterpart per se but rather the interaction of the anthropomorphic appearance and mentalizing stimuli that lead to reactions towards automated counterparts similar to human ones. Unfortunately, a full discussion of the effects of anthropomorphism is beyond the scope of this review. While the selected findings presented here aim to show that appearance seems to matter, we refer to other reviews for a full discussion of the factors involved [see e.g. Złotowski et al., 2015], as well as of the ethical and societal impacts of anthropomorphic machines [see e.g. Darling, 2015].

Summary of section 2.2 In general, studies have shown that interactions with automated agents trigger a reduced emotional and social response in the humans involved. This reduction applies to both sides of the emotional spectrum, and may be beneficial, e.g. where it results in increased economic rationality or higher willingness to disclose sensitive information. It may also be harmful, e.g. where leads to more anti-social or unethical behavior. The intensity of the emotional and social response depends on the characteristics of the human counterpart (e.g. their cultural background) and can be amplified by features of the AA (e.g. its appearance and behavior).

2.3 The importance of task type

Studies show that humans exhibit strong reactions when a previously human-human interaction is transformed into an interaction with an automated agent. An example of this comes from a field experiment which varied whether sales calls for financial services were made by humans or by chat-bots [Luo et al., 2019]. When the caller's identity was not revealed to the customer, chat-bots made about as many sales as experienced human sales workers. However, if the chat-bot's identity was revealed prior to the call, purchase rates fell by almost 80%. As the follow-up survey revealed, the customers claimed that the chat-bots were less knowledgeable and less empathetic - but only if they knew that they were interacting with chat-bots. However, both the identical sales rates and an analysis of the call data showed no performance differences. Similarly, Ishowo-Oloko et al. [2019] in a public goods game find that once participants are aware that their counterpart is a bot, they are less willing to cooperate - with the actual behavior of the counterpart having little impact.

Further studies point to a potential reason for this preference to interact with humans over automated agents: the type of task, and the perceived aptness of using automated agents for such tasks. Indeed, in an experiment investigating the attitudes toward the outsourcing of tasks to robots, people were found to react negatively to the outsourcing of social tasks to robots, but not to the outsourcing of tasks perceived as analytical [Waytz and Norton, 2014]. This distinction was confirmed by several other studies [see e.g. Lee, 2018, Hertz and Wiese, 2019, Castelo, 2019]. Additionally, people tend to react to robots more positively if the appearance and demeanor of a robot match the task it is used for [Goetz et al., 2003, Eyssel and Hegel, 2012]. Such differences in the willingness to interact with automated agents depending on the type of task also appear in studies we will discuss later in the review (see in particular section 3.2 and in particular subsection 4.4).

Several studies considered human attitudes towards using algorithms in creative tasks. In an experiment featuring a poetry competition, judges were unable to reliably tell if a poem was composed by a human or an algorithm - yet rated the human poems higher and stated a stronger preference for human poems [Köbis and Mossink, 2021]. Revealing the nature of the author did not affect the level of aversion exhibited in this study. Similarly, in Jago [2019], respondents rated creative works produced by algorithms as less morally authentic than the identical human works (note that this did not apply to what the authors call 'type authenticity', i.e. the accuracy of the work).

Summary of section 2.3 Studies have documented that people react negatively to automated agents in tasks regarded as social or 'human', but less so or not at all to their involvement in analytical tasks. The *perceived* aptness of automated agents for a task appears to drive these results.

3 Automated agents as peers and superiors in the workplace

The growing capacities and the increasing autonomy of automated agents [Haddadin and Croft, 2016] have vastly augmented the roles they can take in the workplace; from mere tools to be used by humans to increasingly autonomous collaborators or even managers in their own right. As this is a rather recent change, experimental research on how humans interact with such agents in the workplace remains rare. With the advancement of the technology frontier, these interactions are likely to grow in importance. In this section, we therefore discuss some interesting first studies which may help outline a future research agenda. In Section 3.1, we consider studies that look at the effects of automated agents who occupy the role of co-workers and teammates. In turn, Section 3.2 contains studies investigating the use of automated managers. Section 3.3 discusses open questions.

3.1 Automated agents as peers

The introduction of automated agents into existing organizations has so far mostly been investigated using qualitative methods. In particular, a number of observational studies show that the introduction of automated teammates goes beyond a simple replacement process and requires redefining roles, changing existing processes and implies a number of other challenges [see e.g. van den Broek, 2019, Lebovitz et al., 2019, Stubbs et al., 2007].

A first question here is whether humans are willing to use the help of automated teammates, i.e. to outsource some of their tasks to them. Inspired by the real-life experience of Walmart, a company which introduced robots to clean the shops, yet required workers to return to the cleaning tasks if the robots were unable to reach some areas, Alekseev [2020] investigated a setting where parts of a task can be automated, allowing the human workers to focus on the other parts. The study found that the share of people willing to use the automated mode was higher if the automated system could solve a higher share of the tasks. However, on average 30% of subjects chose not to use the automated system - even though it would have led to higher earnings. The main reasons workers gave for not using the automated agent were switching costs and a decrease in being able to focus on the task. This effect was dependent on the individual participants: players with lower cognitive abilities were more likely to opt out of using automation.

Once a hybrid team is formed, a natural starting point for quantitative studies is to investigate the effect of automated agents on the productivity and effort of their human co-workers and teammates. A few studies that consider this question conclusively find that there is a change in human performance, yet disagree on the direction of this effect. In the following discussion, we will attempt to speculate what may trigger the discrepancy in these findings.

In teams where team members execute a very similar task, having an automated teammate seems to decrease the performance of the human teammates [Corgnet et al., 2019, Dell'Acqua et al., 2020]. This is documented, for instance, in a noteworthy lab experiment by Corgnet et al. [2019], where working alongside robots in a conveyor style production task reduces human effort. The authors establish that this reduction in performance is due to the reduced social response towards automated agents- namely, the lack of altruism and social pressure. Similarly, in Dell'Acqua et al. [2020], who consider hybrid teams in a coordination task. The authors demonstrate that even in a task where AI outperforms humans (and therefore should improve the team outcome), the performance of hybrid teams decreases in comparison to all-human teams. This decrease in performance is driven by an increase in coordination failures and a decrease in the effort provided by human teammates - though these negative effects are less pronounced for human players of high skill [similar to the findings of Alekseev, 2020]. Thus, the effect of introducing automation on the performance of human teammates may depend on the type of workers in the team. In addition, the authors also report negative performance effects for all-human teams who worked alongside the hybrid team.

Yet, these negative performance effects might be mitigated by the human teammates' perception of who is responsible to complete a task. Two different studies consider a workplace interaction where the human and automated teammates perform somewhat different tasks, and find that automated teammates increase the effort humans put into the task. First, in a field experiment, Paravisini and Schoar [2013] find that credit officers extend more effort, i.e. spend more time deliberating a decision, when they expect an input (in the form of a score) from a decision-making support system. Similarly, when measuring task distribution in teams in the lab, Gombolay et al. [2015] report that human teammates keep a larger share of the work to themselves when working with robot teammates. A possible explanation for the increase in effort in Paravisini and Schoar [2013] and Gombolay et al. [2015] could be that the human subjects perceived themselves to be more responsible for the outcome, felt that they could not fully rely on the automatic teammate and hence increased their own efforts.

Summary of section 3.1 (part 1) The experimental evidence shows that the introduction of automated teammates can affect the performance of human teammates both negatively and positively. Performance decreases may be due to reduced social reactions to automated agents and/or to coordination difficulties. Performance increases may be due to higher perceived responsibility of the human for a task. Differences in the cognitive abilities of the human partners are important for the willingness to engage and the success of interacting with automated agents.

A closely related question is what effect automated teammates have on the distribution of responsibility within a team. While the diffusion of responsibility in teams is a well documented phenomenon for human groups [see e.g., Darley and Latané, 1968], and one could hence expect that human players might readily shift responsibility to automated agents, in particular given the reduced social concerns against them, this does not seem to be the case. In a laboratory experiment using a modified dictator game, Kirchkamp and Strobel [2019] found no significant differences in taking responsibility between human-human and human-computer teams - though participants in human-computer teams did behave slightly more selfishly. That is, in these types of interactions the decreased social response seems to be balanced out by the lack of agency of automated agents. These results are in line with previous findings by Kim and Hinds [2006], who highlight that the level of autonomy of the robot is important to how much credit or blame is attributed to it. In a recent online experiment, Hohenstein and Jung [2020] considered if AI can be blamed for unsuccessful coordination between two human counterparts by having subjects chat with or without AI assistance (in the form of smart replies by the messenger program). If the team managed to coordinate, the human participants perceived the responsibility for success to be on them, but if it failed, some of the responsibility (ca. 10%) was attributed to the AI system. Interestingly however, participants did not use AI to deflect blame from themselves but rather to reduce the share of blame that they assigned to the partner.

By refraining from shifting (more) responsibility towards computers, the participants forewent a seemingly effective way to avoid being held responsible themselves, as an earlier study showed: Observers of accidents tended to attribute less responsibility to a company if technology was involved in the accident [Kurtzberg and Naquin, 2004]. Similarly, in another study observers attributed less fault to a doctor who followed the (bad) recommendation of an automated decision aid [Pezzo and Pezzo, 2006]. Similar results can be found in a recent study by Liu and Du [2021]. Two recent studies focus particularly on responsibility attribution in moral tasks; Feier et al. [2021] focus on responsibility attribution in moral tasks, and find that in case of an unfavourable outcome, observers assign less blame to the delegator of a moral task if the task was delegated to a machine (rather than to another human). Delegators in the study are not aware that delegating to the algorithms is an effective way to shift blame. Shank et al. [2019] considers the perception of moral violations produced by human-algorithm pairs, in two versions where a human either monitors an AI system or receives a recommendation by it. The authors find that the humans are held responsible for the outcome despite the involvement of AI, yet that humans who monitor AI are faulted less than humans who handle the task alone and than humans receiving AI recommendations. Further studies on this seem necessary, in particular to investigate why observers seem more ready to shift responsibility to automated teammates than to human teammates, and why teammates themselves do not seem to use automated agents as scapegoats even though this seems like an efficient way to deflect responsibility.

Summary of section 3.1 (part 2) Experimental studies have shown that teammates in hybrid teams do not appear to shift blame to their automated counterparts. However, third party observers seem to fault humans less if an automated agent is involved. The autonomy of the automated agent matters for how blame is attributed.

Finally, experiments show that having an automated teammate may affect the interaction among the humans in the team [Traeger et al., 2020, Short and Mataric, 2017, Strohkorb et al., 2016]. For example, one study found that the human teammates were more likely to engage in social conversations and perceived the group more positively if the robotic teammates expressed vulnerability (e.g. by admitting mistakes) [Traeger et al., 2020]. Robots might thus be able to help people collaborate better. However, these studies did not compare the reactions triggered by the robots to those that could be triggered by similarly acting humans in the group. As Rahwan et al. [2020] point out, the strategies that were successfully employed by automated agents in this study are unlikely to work for a human. Based on the finding that automated agents elicit less social responses than humans (see section 2.2), it might be the case that the social cohesion triggered by robots is lower than the one that would be triggered by a human.

3.2 Automated agents as superiors

Today, an increasing number of managerial tasks are performed by automated agents [see e.g. Aharoni and Fridlund, 2007, Lee, 2018]. Unsurprisingly, this has started to receive attention in behavioral research, and there is some first experimental evidence on whether and when humans are willing to accept automated managerial decisions.

First, experiments have documented instances where people are willing to accept automated managers. Cormier et al. [2013] demonstrated that people are willing to obey automated managers: In an obedience experiment, the manager instructing participants to complete a tedious task is either a human or a robot. While participants protested significantly more frequently, earlier and quit the task sooner when instructed by a robot, they were still willing to obey it. Indeed, about half of the participants in this condition continued with the tedious task for the full duration of the study, even after trying to protest. In a field experiment with Alibaba warehouse workers, a substantial share of participants were even found to prefer automated managers to human ones [Bai et al., 2020]. Here, warehouse workers received lists of items to collect from shelves either from a human manager or from a computer terminal. The workers in the experiment generally perceived the lists distributed by an algorithm through the computer terminal to be fairer than those handed out by human managers (although both were generated by the same algorithm). As a result of differences in perceived fairness of task assignment, warehouse productivity increased in the automated conditions. A lab experiment using a lego set assembly task further documented that people are content to have a robot allocating tasks within the team, and may even assign them such managerial responsibilities themselves [Gombolay et al., 2015]. Additionally, this study finds that both subjective and objective measures of participants' satisfaction increased with increased autonomy of the automated agent. Workers preferred to have the robot make scheduling decisions, and spent less time on rescheduling tasks the more autonomous the robot was.

However, note that another experiment in a similar setting finds the opposite result, namely that participants disliked robotic managers, were more likely to blame them for mistakes, and relied upon them less [Hinds et al., 2004]. Importantly, while in Gombolay et al. [2015] the robots actually performed managerial tasks (e.g. scheduling), in Hinds et al. [2004] they performed supporting tasks within the team (e.g., carrying the parts), and were exogenously assigned managerial status. Hence, as speculated by the authors in Hinds et al. [2004], the aversion to robot managers may be due to a mismatch in skills and authority, not due to the robotic nature of the supervisor itself.

The acceptance of automated managerial decisions might also depend on the context of the task. Accordingly, Lee [2018] finds that human and automated managerial decisions are perceived as equally fair for analytical tasks - but not for social tasks, where the human decisions are perceived as fairer. Similarly, Newman et al. [2020] also document that the use of AI for more social managerial tasks might not be very well accepted. In a series of vignette experiments, the authors consider how employees perceive a variety of HR decisions (layoffs and promotions, performance review, determining the size of the bonus, evaluating interviews) if they are made by an algorithm. In contrast to the findings of Bai et al. [2020], their results document that people perceive algorithmic decisions as less fair, consider them to be reductionistic, leading them to think that certain qualitative information or contextualization is not being taken into account.

Finally, with regards to automated performance evaluation and bonus assignment, Strobel [2019] finds that employees exert significantly more effort if the threshold performance for receiving a bonus is set by a human manager manually on case by case basis as compared to setting the threshold ex ante, before performance of individual employees is known and passing on to an automatic system. However, this result appears not to be driven by fairness considerations or task context, but by the fact that employees expect a lower performance threshold in the automated condition (i.e., expect to receive a bonus even with lower effort).

Summary of section 3.2 The experimental evidence shows that humans can accept automated agents performing managerial tasks, and that in some contexts automated managers may even be preferred over human ones. An important factor driving the reactions to automated managers is the type of the task (analytical vs. social).

3.3 Open questions

In summary, the cited experiments on human-machine interaction in the workplace show that people behave differently in the presence of automated co-workers or managers, and that there may even be spillovers into the interactions between humans on the team. It seems that there are several behavioral channels through which the introduction of an automated teammate may negatively affect the performance of human co-workers. Yet, the perception of responsibility for the task appears to be a channel that might mitigate or even override the negative effect. As for automated managers, it appears that the nature of the managerial task plays a particularly crucial role in the reactions to automated managers.

Given that managerial tasks carry special importance by affecting organisational success at large as well as productivity of individual employees, it seems to us that a particularly promising avenue of research is to investigate where automated or human managers might be particularly suited. Based on the findings discussed in Section 2 and the evidence above, automated managers may be especially suitable for more analytical tasks and have an edge over human ones when it comes to making unpopular decisions, as these might trigger less of an emotional response in the human subordinates. On the other hand, the decreased social response to both automated managers and teammates might carry the problem of increased misreporting and unethical behavior. This concern may be counterbalanced by the appearance and anthropomorphism of the agents. However, this in turn may affect the perceptions of agency and responsibility within the team.

In conclusion, more systematic research is needed in order to understand how we can leverage the peculiarities of human-machine interaction in the workplace. As technology advances rapidly, developing our body of knowledge on the topic of hybrid interaction in the workplace and group dynamics is of high urgency.

4 Automated agents as delegates and aids in decision-making

A topic that has received particular attention in the literature is the use of automated agents in decision-making. The experiments discussed in the following section all contribute to the understanding of whether humans are willing to accept the involvement of automated agents in decision-making - and, if so, to what extent? Are they only willing to accept automated decision aid, or are they willing to fully delegate decisions to them? Yet, although the research questions of the papers in this section are similar, the results often are not. Indeed, as we will discuss in this section, the experiments will document different behavioral patterns, and in consequence, will sometimes come to different conclusions with regards to how humans perceive the use of automation in decision-making: In particular, several studies document that humans are averse to delegating decision tasks to automated agents, a phenomenon called *algorithm aversion*. Other experiments find that humans prefer automated advice to human advice, a phenomenon called *algorithm appreciation*. Yet further studies find that humans over-rely on automated decision-making supports, failing to correct for their mistakes (*automation bias*) or to properly monitor them (*automation-induced complacency*).

At first glance, these results seem puzzling - how can people simultaneously be averse to automated decision-makers, and prefer their advice to human advice? However, as we will discuss in the following sections, there are a number of factors which might explain these patterns in a consistent way. In an attempt to reconcile these seemingly inconsistent findings, we categorize the contexts in which they occur according to the distribution of agency within the interaction.That is, we propose to distinguish between situations in which the primary or exclusive authority over the decision is delegated to the automated agent (Section 4.1), and those in which the human retains the primary/exclusive authority over the decision (Sections 4.2 and 4.3). As we will discuss, the experimental findings suggest that this distinction in agency is of major importance for how humans react to algorithmic decision supports. In the final Section 4.4, we will also discuss further potential factors which might explain the inconsistencies in findings.

4.1 The aversion to automated agents as delegates

Let us first consider a situation which is currently steadily increasing in importance: Decision-making where the authority over the decision is ceded to an automated agent. This cession of authority can be done fully - think, for example, of an automated trading program, which is set up initially by the user, but then makes autonomous sales and acquisitions on the stock market. It can also be done only partially - think, for example, of an algorithm choosing a small selection of candidates to be interviewed by a human employer. In both of these examples, an important part of the decision is delegated to the automated agent. The question which arises here is whether, and under what circumstances, humans are willing to make this delegation.

In an influential series of experiments investigating this question, participants were tasked with a number of forecasting tasks (regarding the future success of MBA graduates and the future development of airline passenger counts), and given the choice between making these predictions themselves or delegating the task to an algorithmic forecaster [Dietvorst et al., 2015]. The participants were aware that the algorithm had made some mistakes in the forecasts - but it still consistently outperformed the human participants. Nevertheless, the participants were more likely to choose to make the prediction themselves. This costly choice is an example of a behavior the authors of the study termed *algorithm aversion*. The preference of humans to rely on themselves rather than to use superior automated decision-making supports had also been documented previously in other contexts, including when driving a car [Kantowitz et al., 1997] or when solving a visual detection task [Dzindolet et al., 2002].

Importantly, algorithm aversion does not simply imply that people prefer to make decisions themselves. Given the choice between receiving advice from a human or an automated agent, people were more likely to go with the human for joke recommendations [Yeomans et al., 2019], for medical tasks [Promberger and Baron, 2006, Longoni et al., 2019] and when the task was unknown [Hertz and Wiese, 2019]. Doctors were perceived as less qualified by observers if they delegated decisions to automated decision support, but not when they sought advice from human colleagues [Arkes et al., 2007, Shaffer et al., 2013].

Algorithm aversion may affect decision-making: In experiments, participants were more likely to follow medical advice when it came from a human provider rather than from an AI agent [Longoni et al., 2019], and gave more weight to investment advice by humans rather than a statistical forecast [Önkal et al., 2009]. In an observational study, managers at a clothing retailer were prone to reject a decision support system's advice regarding markdowns for sales [Saez de Tejada Cuenca, 2019]. And, indeed, the debate about when people would do better if they followed algorithmic advice dates back decades [Meehl, 1957, 1954].

Multiple possible causes of algorithm aversion, or more generally of the "disuse, or the neglect or underutilization of automation" [Parasuraman and Riley, 1997], have been proposed by the literature. A first candidate is low trust in automated agents. Indeed, people were found to be particularly unlikely to use automated decision-making aids in uncertain domains [Dietvorst and Bharti, 2019], where trust is arguably more important. In the study on the use of automated investment advice, participants stated that they trusted automated agents (slightly) less - though this did not predict whether they actually followed the automated advice [Önkal et al., 2009]. A potential reason for the lack of trust in automation is advanced by B. Dietvorst and coauthors [Dietvorst et al., 2015, 2018, Dietvorst and Bharti, 2019], who argue that the cause of algorithm aversion is seeing an algorithm err. According to this argument, while people are willing to forgive humans for making mistakes, this leniency is not extended to algorithms. Indeed, earlier studies also show that while participants initially trusted automated agents more than humans and preferred to use them, they lose trust quickly and prefer human aids after seeing the automated agents make mistakes [Dzindolet et al., 2002]. Machine errors have been shown to cause low trust in their performance, particularly if error rates are not constant [Muir and Moray, 1996]. Particularly strong support for the argument of the loss of trust after seeing an algorithm err comes from a more recent study, which finds no support for the hypothesis that human advice is generally preferred to automated advice, but shows that after receiving incorrect advice the utilization of automated advice decreases significantly more than the utilization of human advice [Prahl and Van Swol, 2017].

However, findings from other studies cast doubt on these arguments. For instance, the resistance to the use of an AI health provider documented in a series of experiments persisted even when the AI agent was specifically described as being superior to the human regarding the number of complications/accurate diagnoses [Longoni et al., 2019, study 3c]. Neither the (insignificant) differences in trust extended towards automated agents or humans nor the perceived utility of the agent were able to explain the preference for the human decision-maker documented in another experiment [Gogoll and Uhl, 2018]. And, two studies of social robots even found that people did either not mind the robots' mistakes [Salem et al., 2015], or even preferred robots who made mistakes to those who performed flawlessly [Mirnig et al., 2017]. Note that, presumably, these last findings were related to the task context: While the above mentioned studies which document salience of machine errors mostly used analytical tasks [e.g., Dietvorst et al., 2015, Önkal et al., 2009], these robots were used for social tasks. It seems possible that there is some kind of interaction effect between task characteristics and sensitivity to errors: Humans may not be willing to accept automated mistakes in analytical contexts, but they seem to accept or even prefer fallible machines in more social contexts - after all, it would only make the machines appear more human. More generally, the findings of automation bias and algorithm appreciation to be discussed in Sections 4.3 & 4.2 show that people do not by and large distrust machines. Indeed, as we will discuss, these papers document instances where humans trust automated agents too much. Trust might also depend on the anthropomorphic appearance of the automated agent (for an example with autonomous vehicles see Waytz et al. [2014], for a general overview of the topic of trust in machines see e.g. Lee and See [2004], Glikson and Woolley [2020]).

A second potential cause of algorithm aversion, which is advanced by another experiment [Gogoll and Uhl, 2018], is the moral context of the decision. The authors define a moral decision as a decision which affects a third party. In this experiment, participants are given the choice between delegating a calculation task to another human or to an algorithm. The earnings for the task are given to a third person, placing the task in the moral domain. Participants were not only significantly more likely to go with the human decision-maker, but even punished others who chose the algorithm. As neither the trust extended towards the agent (measured using a trust game), nor its perceived utility can explain these findings, the authors argue that people might exhibit a *per se* aversion to the use of automated decision-making agents in the moral domain (as stated, defined as decisions which may affect third parties, e.g. decisions by automated driving systems). This is corroborated by another series of experiments which also documents that humans are averse to automated agents making moral decisions ("morally-relevant driving, legal, medical, and military decisions", [Bigman and Gray, 2018, p. 21]). Interestingly, this aversion holds irrespective of whether the decisions made are favorable for the third party affected by them. A potential insight into the reasons behind this preference comes from Longoni et al. [2019]: Here, the authors propose that the cause of algorithm aversion is "uniqueness neglect", i.e. the concern that an automated agent is unable to account for an individual's unique characteristics and circumstances. Testing this hypothesis in the healthcare context, the authors find that a participant's perceived sense of uniqueness is able to predict their aversion to the use of medical AI. Hence, these studies show that the moral context of a decision might be the factor driving algorithm aversion. However, this explanation does not seem satisfactory for the aversion documented in the studies by B. Dietvorst and coauthors: Here, algorithms are used to forecast the success of MBA graduates and the number of airline passengers [Dietvorst et al., 2015], standardized test scores [Dietvorst et al., 2018] and for a number of deliberately abstract judgement tasks [Dietvorst and Bharti, 2019] - all tasks outside of the moral domain.

However, we feel that there is another important factor to note. In almost all the studies we cited above, the participants are found to exhibit algorithm aversion when they are given the choice between a human *or* an automated decision-maker [Dietvorst et al., 2015, Gogoll and Uhl, 2018, Promberger and Baron, 2006, Yeomans et al., 2019, Hertz and Wiese, 2019, Önkal et al., 2009, is a notable exception]. In other words, algorithm aversion seems to occur where humans cede the authority to make a decision to an automated agent. However, a different situation arises when an automated decision-making system is used not to replace the human decision-maker, but to assist him. As we will discuss in the following two sections, there is evidence which suggests that this factor may not only remedy algorithm aversion, but even trigger a positive preference of automated over human decision-making support.

Summary of section 4.1 Experimental studies show that when able to delegate a decision task to an automated agent, people frequently exhibit *algorithm aversion*, i.e. prefer to make the decision themselves or to delegate it to another human. Some studies link this aversion to lower trust and lower performance expectations in AA compared to humans. However, algorithm aversion has also been documented in cases where the AA has and is perceived to have superior performance and where there seems to be no difference in trust extended. Algorithm aversion seems to be particularly pronounced in moral contexts, and might be caused by the perception that machines are unable to account for human individuality. As we will discuss in the next section, if AA is in a supporting role to a human, some studies find the opposite, i.e. document instances where automated decision support is preferred over human support.

4.2 The over-reliance on automated decision aids

We now consider a different kind of the use of automated agents in decision-making: to assist and support, but not to replace the human decision-maker. Examples of such settings could be a pilot flying a plane with the help of an autopilot, the operator of a machine being assisted by an automated alert system, or a person drafting a text being supported by a spell-checking software. A series of experiments, mostly from psychology or applied engineering fields, have documented that people frequently over-rely on the automated agent's advice in these settings, fail to realize when the automated advice is wrong and fail to intervene when they should [for reviews of this literature see e.g. Parasuraman and Manzey, 2010, Alberdi et al., 2009]. This phenomenon is termed *automation bias*, or the over-reliance on automated decision-making support. Note that, in fact, this literature describes two related phenomena: automation bias, i.e. the failure to intervene, and *automation-induced complacency*, i.e. the failure to appropriately monitor automated support. We will refer to both of these phenomena using the broader term of automation bias [following Parasuraman and Manzey, 2010].

Automation bias occurs when people treat the automated agent "as a heuristic replacement for vigilant information seeking and processing" [Mosier and Skitka, 1996, p.205]. It manifests itself in errors of commission, i.e. misguided actions based on a false alarm by the automated system, or errors of omission, i.e. the missing of critical events when the automated system fails to flag them [Skitka et al., 1999]. In one of the earliest studies of this phenomenon, pilots were found to be more easily misled by (faulty) automated checklists than by simple paper checklists [Mosier et al., 1992]. This result was then replicated with a flight simulation task in the lab [Skitka et al., 1999]. Here, participants in automated conditions were more likely to miss events not flagged by the software (i.e. to commit errors of omission), and more likely to act on incorrect software advice (errors of commission) compared to flying under manual control. Both laypeople and experts are susceptible to automation bias, as was documented in a number of flight simulation studies with experienced pilots [Mosier et al., 1998, Sarter and Schroeder, 2001, Galster et al., 2001]. Similarly, in a study with experienced air traffic controllers, significantly fewer controllers detected a conflict that the system had failed to flag compared to when conflicts were handled manually [Metzger and Parasuraman, 2005].

Later studies document the over-reliance on automated decision support outside of the domain of aviation. For instance, the use of a spellcheck program was shown to prompt people with high writing skills to miss spelling errors and to falsely change correct spellings [Galletta et al., 2005]. In health care, the reliance on incorrect automated advice was shown to lower the decision accuracy of physicians reading EKG charts [Tsai et al., 2003]. The sensitivity of professional film readers reading mammograms decreased if they used computer-aided detection systems - causing them to miss cancers if the system had failed to mark them [Alberdi et al., 2004]. This result was replicated in a follow-on study which in addition found that non-cancerous objects were more likely to be marked as cancers if the system had falsely flagged them as such [Alberdi et al., 2008]. Further, automation bias and automation-induced complacency have also been documented in process control [Bahner et al., 2008, Manzey et al., 2012, Wickens et al., 2015] and command and control situations [Rovira et al., 2007]. In summary, the evidence on automation bias shows that while automated decision-making aids can improve decision quality when they are correct, they may also lower decision quality when they are incorrect, because people are nevertheless prone to rely on them.

Automation bias seems to be more likely to occur in situations of high cognitive load, which may stem from task complexity, multitasking or from time pressure [Parasuraman et al., 1993]. Notably, experience with automated systems or with sharing the load with teammates do not seem to remedy automation bias [Skitka et al., 2000b, Mosier et al., 2001]. Indeed, automation bias seems not to stem from general inattention, but from the monitoring of the automated agent having lower priority and receiving less attention than other competing tasks [Parasuraman and Manzey, 2010]. In line with this, an experiment has documented that providing people with variable priority attention training reduces automation bias [Metzger et al., 2000]. A further effective way to reduce automation bias seems to be highlighting the responsibility and accountability of the monitoring person [Goddard et al., 2012, Skitka et al., 2000a].

Summary of section 4.2 Experiments have documented that when people are supported by an automated agent in a decision task, they are likely to exhibit *automation bias*, i.e. to over-rely on the automated support and to (fail to) intervene following its incorrect inputs. Automation bias may occur irrespective of the level of experience of the user. Studies show it to be more likely in situations of high cognitive load. Automation bias can be mitigated by prompting higher priority to monitoring the system by i.e., highlighting the responsibility and accountability of the human, or by training attention capabilities.

4.3 The preference for automated decision aids

The decision-making situation in which a human is assisted by an automated agent has also started to receive attention in the behavioral economic literature. However, the focus of these studies is slightly different to the ones investigating automation bias: here, the question asked is not how a human will behave when assisted by an automated agent, but whether humans want such automated assistants. For example, think of a physician who can choose to use an automated diagnostic support system to help diagnose patients, or of a person asked to make a prediction and given the option of receiving automated support for this. That is, in contrast to the settings of the experiment documenting algorithm aversion discussed in section 4.1 and as in the experiments discussed in section 4.2, here the automated agent does not replace the human decision-maker, but assists them. The few economic studies which investigate this situation have shown that, in this set-up, humans are appreciative of automated advice and may even prefer it to human advice - a phenomenon termed *algorithm appreciation*.

In the leading series of experiments on algorithm appreciation, Logg et al. [2019], who also coined the term, ask participants to make a number of quantitative judgments (visual estimation of a person's weight, prediction of physical attractiveness, prediction of song popularity) under uncertainty, and are offered advice from either humans or automated agents [Logg et al., 2019]. The participants consistently chose to receive and put more weight on automated advice, rather than human advice. Hence, the authors conclude, people seem appreciative of automated decision-making support. Further empirical support for this appreciation of automated advice is found in an earlier experiment by Dijkstra [1999]: Here, participants are provided with (correct) human advice and (incorrect) algorithmic advice for a legal case. The participants very frequently relied on the algorithmic advice, in particular when it was given in production rule form (the latter replicating a previous finding from Dijkstra et al. [1998]). Outside of the lab, people were found to be responsive to algorithmic advice when choosing health care plans in a randomized control trial [however, this study does not feature a comparison to the responsiveness to human advice, Bundorf et al., 2019].

Interestingly, a close look at the leading experiment on algorithm aversion by Dietvorst et al. [2015] shows that participants in the control condition, who did not see the model perform, also stated that they were more confident in the model's, rather than in the human, forecasts. More generally, the finding that humans appreciate algorithmic advice is supported by a number of results contained in studies which generally document algorithm aversion, but find that this aversion is decreased or even eliminated when the agent's role is framed as only advisory. Indeed, for example, in Dietvorst et al. [2018], people are more likely to use a forecasting algorithm when they are able to intervene and even just slightly modify its advice. Other studies show that participants are less averse to the use of automated agents in medical and other moral decisions when the agent provides the humans with a recommendation, rather than a decision [Promberger and Baron, 2006, Bigman and Gray, 2018]. Similarly, a vignette study in a consumer and medical setting finds that participants appreciate professionals' use of automated decision aids, but react negatively to a full delegation of decision authority [Palmeira and Spassova, 2015]. Finally, framing a medical AI system as providing support rather than replacing the human health care provider fully eliminates the algorithm aversion documented in the rest of a series of experiments in [Longoni et al., 2019, study 9].

Summary of section 4.3 Experimental studies have shown that, when people are assisted in making a decision by an automated agent, they can exhibit *algorithm appreciation*, i.e. be appreciative of automated advice and may even prefer it to human advice. The shift from algorithm aversion (see section 4.1) to algorithm appreciation seems to occur when people are allowed to intervene and adjust the recommendation of the automated agent.

4.4 Open questions

In summary, while the studies discussed in section 4.1 document instances where humans are averse to the use of decision-making algorithms, the studies discussed in sections 4.2 and 4.3 document instances where humans are indifferent between human or automated decision-makers, or even prefer and/or over-rely on automated rather than human decision advice. What emerges is that there is a clear need for future, integrative studies which investigate the factors triggering one or the other phenomenon. In the following we deliberate on a few hypotheses that such studies could investigate.

First, as we have already alluded to, a factor which could trigger algorithm aversion in one case, and algorithm appreciation or automation bias in the other, is the distribution of agency in the interaction. There is evidence that people seem particularly appreciative of automated agents when they are framed as providing support rather than as independently making the decision. To cite but one example, in one of the vignettes of a study that generally documents the aversion of people to machines making moral decisions, participants are asked to choose who they would prefer to make a moral decision: a human doctor, an "autonomous statistics-based computer system", or the doctor advised by the automated system [Bigman and Gray, 2018, p. 26]. The participants were most likely to choose the last option. In the words of the authors, "[t]hese results suggest that most people are willing to have machines involved in moral decisions, as long as they are not the ones to make the actual decisions" [Bigman and Gray, 2018, p. 30]. More generally, the empirical evidence seems to suggest that humans are averse to fully giving up their decision authority, yet appreciative of automated advice when they retain - or when they feel that they retain - the ultimate authority over the decision. However, this hypothesis so far rests on a small number of studies and some corollary results, while studies which purposefully investigate the impact of this factor remain lacking. Hence, it remains unclear whether the exact degree of human involvement matters - does the human need to retain real determinative capacity, or is it sufficient if humans feel like they remain involved? And, if such nominative human involvement were sufficient to avoid algorithm aversion, would it be ethically and/or legally defensible to mislead people into thinking that they remain involved in decision-making?

A second factor that might explain the discrepancy is the context of the decision, respectively the type of task contained in the decision. In particular, the empirical evidence we have discussed suggests that while humans are willing to accept automated agents in areas considered more objective or analytical, they are more reluctant to do so in areas considered as social or moral.

This could be due to normative preferences, i.e. the view that some decisions simply should be made by humans [see e.g. Gogoll and Uhl, 2018, who speak of a *per se* aversion to the use of algorithms in the moral domain]. On the other hand, it could also be due to performance expectations, i.e. the view that humans are simply better at making social or moral decisions. A formulation of this argument comes from the study by Longoni et al. [2019], who argue that 'uniqueness neglect', the view that automated agents are unable to account for the uniqueness of humans, is the cause of algorithm aversion. In other words, people might be averse to the use of decision-making algorithms in social or moral situations because they expect them to perform badly in these decisions (for more on task context see also section 2.3).

Third, performance expectations, or the perception of the relative capabilities of the human and the automated agent may trigger algorithm aversion or appreciation irrespective of the context - and irrespective of whether this perception is accurate or not. An illustration of this comes from Logg et al. [2019], who replicate the airline passenger forecasting task first used in Dietvorst et al. [2015]. The replication shows that when participants choose between themselves and the algorithm, they prefer to do the task themselves [as was the case in Dietvorst et al., 2015]. In contrast, when participants choose between another human and the algorithm, a substantial part of them exhibits algorithm appreciation [Logg et al., 2019]. Hence, Logg et al. [2019] argue, (over-)confidence in one's own abilities may lead participants to exhibit algorithm aversion, yet doubts in the abilities of others may lead them to exhibit algorithm appreciation. This view is corroborated by a second difference between the two studies: in Dietvorst et al. [2015] participants have seen the algorithm make mistakes, while participants in Logg et al. [2019] received no information on performance prior to making their decision. Finally, the importance of perceived capabilities seems mirrored in the finding that experts are more likely to discard automated advice than are non-experts [see e.g. Logg et al., 2019, experiment 4; Tazelaar and Snijders, 2004]. Some additional evidence for the importance of performance expectations on the willingness to accept automated decision aids comes from Efendić et al. [2020], who in a series of experiments document that humans are more willing to use human advice if it is given slowly, yet that the opposite applies to algorithmic advice: here, the participants were less likely to follow the advice if response time is higher. Interestingly to the present discussion, in one of these studies the authors, while focusing exclusively on automated advice, document that people state that would be more likely to disregard the automated advice and consult with a human colleague if response time is higher [study 6 in Efendić et al., 2020]. Finally, while there is no comparison to accepting human advice, Filiz et al. [2021] show that when participants receive feedback and are hence able to learn that the automated agent gives high-quality advice, the willingness to use its advice increases.

What emerges is that there is a clear need for more integrative studies to investigate which factors trigger the appreciation of, or aversion to, automated agents involved in decision-making situations. Without such studies, researchers run the risk of focusing only on one kind of behavior, and extrapolating it to too many situations, as is nicely illustrated by a survey contained in Logg et al. [2019]: When the authors asked researchers in the field to predict how people would behave in their experiment, most researchers predicted that the participants would display algorithm aversion - yet the opposite happened. We hope that the hypotheses we have listed here could provide a good starting point for such research on the factors which influence the attitude of humans towards automated agents involved in decision-making.

5 Discussion and directions for future research

The behavioral research reviewed in this article provides a number of important insights into how humans interact with automated agents. However, many open questions remain, as sections 3.3 and 4.4 have pointed out in detail. The importance of finding answers to these questions goes beyond a purely academic pursuit of knowledge. From an organizational standpoint, it is vital for firms to know how humans might interact with automated agents in various contexts. From a political standpoint, the recent technological advances have caused extensive discussions of the risks and benefits of automation [see e.g. Pasquale, 2015, Cowgill and Tucker, 2019]. These discussions have been accompanied by repeated calls for regulation [see e.g. Citron and Pasquale, 2014, Doshi-Velez et al., 2017, Wachter and Mittelstadt, 2019]. Both public opinion [Grzymek and Puntschuh, 2019] and policy-makers seem to support stricter regulation of automation, in particular with regards to AI [see e.g. Wallace, 2020, for the EU, Roberts et al., 2019, for China, and OECD, 2019, for the OECD]. The empirical research reviewed in this article may provide a number of insights for the development of such regulation, and for the development of optimal firm practices in an age of automation - as the following paragraphs will show with a number of examples.

Areas of application First, while the ever-moving technological frontier constantly offers new areas for automation, the research reviewed here shows that the question of where automation is beneficial cannot be answered solely on the basis of technical considerations. Indeed, the evidence shows that interacting with automated agents reduces the emotional and social response of humans. Hence, where social rules or emotions form obstacles, automation may bring benefits beyond those of time and labour savings - for example, by increasing the willingness people to disclose sensitive information like intimate partner violence [Ahmad et al., 2009, Humphreys et al., 2011]. In contrast, in other situations social rules and expectations are desirable, and automation may prove detrimental. For example, studies on charitable giving have repeatedly shown that social image concerns and social pressure are important drivers of charitable giving [DellaVigna et al., 2012, Andreoni et al., 2017] - and one may hence speculate that automated solicitors might gather fewer funds than human ones. In an organisational context, due to the high importance of peer effects for worker performance [Mas and Moretti, 2009], replacing human peers with automated ones might decrease team performance.

Degree of automation A second insight concerns the optimal degree of automation. Numerous studies have documented that fully automated decisions can be more accurate and less discriminatory than human decisions [see e.g. Saez de Tejada Cuenca, 2019, Kleinberg et al., 2018, Stevenson, 2018, Hoffman et al., 2018, Gates et al., 2002]. Hence, one might argue, it would be beneficial to take humans off the decision loop, thereby simplifying the decision problem to the task of creating the most accurate and least biased algorithm [see e.g. Miller, 2018]. However, the studies reviewed here show that humans are particularly likely to exhibit algorithm aversion when they are - or when they feel that they are - replaced by automated agents. Entirely removing humans from the decision loop could thus make them particularly likely to mistrust and make inefficiently little use of algorithms. This empirical finding seems to be reflected in official government positions [see e.g. Hancock, 2015] and even in regulation [see e.g. art. 22(1) EU GDPR]. However, further studies reviewed in this article also point to a potential downside of retaining human involvement in automated decision-making: people may over-rely on the recommendations produced by automated support systems, and fail to correct for the system's mistakes. For the moment, the reviewed evidence shows that the degree of human involvement constitutes an important factor which affects people's reaction towards automation. Further studies investigating this phenomenon will be necessary to solve the problem of the optimal degree of automation.

Accountability A particularly promising avenue of research in this regard seems to be the investigation of the impact of (real or perceived) accountability. Indeed, studies suggest that highlighting the responsibility of a human agent for the outcome of a decision could both decrease algorithm aversion by way of ensuring human intervention [see e.g. Dietvorst et al., 2018] and decrease automation bias by ensuring more effective monitoring [see e.g. Goddard et al., 2012]. However, ensuring human accountability has recently become more difficult with the growing use of opaque, 'black-box' algorithms [see e.g. Pasquale, 2015, Lipton, 2018].

Transparency The obvious solution to this problem seems to lie in increased algorithmic transparency. Behavioral research can again provide valuable insights here. For instance, Erlei et al. [2020] shows that while transparency seems to contribute to the willingness to use a system in general, users seem be less likely to incorporate the suggestions of a more transparent system into their actions [Erlei et al., 2020]. Moreover, experiments show that transparency can also backfire; too much transparency can discourage people from using technology [Kizilcec, 2016, Lee et al., 2019], or create incentives to game the system [Ederer et al., 2018]. In addition, transparency can be difficult to establish and have unforeseen effects. To illustrate this point, consider a particularly interesting finding from a computer science user study which develops an explanation method for a machine learning algorithm used to distinguish pictures of wolves and huskies [Ribeiro et al., 2016]. To test the explanations, the authors first showed a group of students trained in machine learning a number of classifications the algorithm had made. When the students were shown the raw data (the pictures) as well as the labels the algorithm had proposed, most of them stated that they trusted the algorithm to perform well, even though they could see that it had made some mistakes. Only if, in addition, the authors showed the students an explanation of how the algorithm made these classifications (namely, that the snow in the background of the picture was a relevant factor) did the students lose trust in the algorithm - even though they had already known its accuracy rate before receiving the explanation. Hence, explanations matter: simply providing information about the accuracy of algorithm is not sufficient to enable observers to accurately judge an algorithm's validity. Indeed, a recent experiment shows that the way an algorithm is explained seems to affect how people perceive it [Cowgill et al., 2020].

More generally, transparency about automated decision-making can be misleading if there is no appropriate comparison to human decision-making. An example of this comes from a study which compares a CV screening algorithm and human HR managers [Cowgill, 2018]. First, the author establishes that the algorithm gives a negative weight to candidates from non-elite universities. Thus, one might conclude, the algorithm is biased and harmful. However, it turned out that the candidates from non-elite schools actually disproportionally benefited from being screened by the algorithm, as the human evaluators would have assessed their credentials even more negatively. A similar finding comes from another study which finds that an automated recidivism risk score is biased, but not that the involvement of human decision-makers would result in any less biased decisions [Tan et al., 2018].

Perceptions of algorithmic decisions On top of the facts of how unbiased decisions are, perceptions of those decisions by people are important. People may perceive algorithmic decisions as less than fair even when they objectively are [Lee and Baykal, 2017] or think that automated decisions are more fair than identical human decisions [Bai et al., 2020]. Further research will be needed to fully establish the impacts of transparency on automated decision-making.

6 Conclusion

The experimental evidence reviewed in this article shows that human-automated agent interactions are different from human-human interactions. In general, these interactions trigger less emotions, and are less affected - but not unaffected - by social concerns. People seem more willing to interact with automated agents in contexts they perceive as analytical or objective - this seems to apply both to their attitude in interacting with automated agents and to their willingness to use automated decision-making support. In the workplace, humans have started interacting with automated agents who take the roles of team-mates and managers. While the existing evidence clearly shows that the introduction of automated agents into the roles of team-mates and managers has an effect on the behavior of humans in the workplace via a number of different channels, more careful consideration of when, for what tasks and with what type of human workers automated agents interact seems necessary before behavioral regularities and ensuing policy advice can be formulated. Another area in which many open questions remain is automated decision-making: while some experiments find that people prefer and (over-)rely on automated advice, rather than human advice, others find that they are averse to automated decision-makers replacing human ones. We have discussed a number of factors which might drive these results, including the distribution of agency in the decision-making situation, the context of the decisions and performance expectations. In summary, the reviewed studies not only show what we know about human-automated agent interactions, but also point out a number of open questions and avenues for future research.

Our hope is that this review and discussion can serve to inform decision-makers and managers who consider using automated agents in their practices. With the evidence gathered here, and by highlighting potential inconsistencies and avenues for future research, we aim to further stimulate the lively research on this topic across disciplinary boundaries.

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A Appendix: Methodological overview of reviewed studies

The below tables provide details on the studies reviewed in this article. Table 1 gives short methodological notes on the experimental studies. Table 2 briefly describes the mentioned observational studies. Table 3 briefly describes the mentioned literature reviews.

The following should be noted about the classifications used in table 1: Where a study contains multiple experiments, experiments are enumerated following the authors' system. Wherever authors do not explicitly give information about a methodological detail about their study, the table states "N/A". In contrast, the use of "None", means that the authors state that a method was not used. E.g., regarding incentives, "None" signifies that no incentives were used, whereas "N/A" signifies that the paper does not specify either way. Incentives are differentiated into contingent or non-contingent incentives, without further elaboration of whether they consist of monetary payments or non-monetary rewards (such as e.g. course credits). Further, regarding the elicitation measures, the observation of participants' actions and choices is classified as revealed preferences independently of whether they are incentivized or not.

If possible, in the column "Title and Disciplene(s)" we additionally report the discipline(s) of the journal which published the article according to Scimago Journal classification by Scopus. It classifies articles into 27 broad fields with 2-digit-codes. The classification is not available and thus omitted for working papers, books and dissertations. Disciplines are coded as follows: 1-Arts and Humanities, 2-Psychology, 3-Economics, Econometrics and Finance, 4-Business, Management and Accounting, 5-Decision Sciences, 6-Social Sciences, 7-Computer Science, 8-Medicine, 9-Engineering, 10-Neuroscience, 11-Multidisciplinary, 12-Nursing, 13-Health Professions, and 14-Mathematics.

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Adam, Krämer, et al., 2015	"Auction fever! How time pressure and social competition affect bidders' arousal and bids in retail auctions" (4)	1: Lab	240	Students	Contingent	Revealed preferences & phsyiological measures
		2: Lab	216	Students	Contingent	Stated, revealed preferences
Adam, Teubner, et al., 2018	"No rage against the machine: how computer agents mitigate human emotional processes in electronic negotiations" (1, 4, 5, 6)	Lab	216	Students	Contingent	Revealed preferences & physiological measures
Aharoni and Fridlund, 2007	"Social reactions toward people vs. computers: how mere lables shape interactions" (1, 2, 7)	Online	40	Students	Non-contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Ahmad et al., 2009	"Computer-assisted screening for intimate partner violence and control: a randomized trial" (8)	RCT	144 (treat-ment) + 149 (control)	Adult women	N/A	Stated, revealed preferences.
Alberdi, Povyakalo, Strigini, and Ayton, 2004	"Effects of incorrect computer-aided detection (CAD) output on human decision-making in mammography" (8)	1: Lab in the field	20	Clinical experts	N/A	Stated, revealed preferences
		2: Lab in the field	19	Clinical experts	N/A	Stated, revealed preferences
Alberdi, Povyakalo, Strigini, Ayton, and Given-Wilson, 2008	"CAD in mammography: lesion-level versus case-level analysis of the effects of prompts on human decisions" (7, 8, 9)	Lab in the field	50	Clinical experts	N/A	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Alekseev, 2020	"The Economics of Babysitting a Robot"	Lab	128	Students	Contingent	Revealed preferences
Arkes et al., 2007	"Patients derogate physicians who use a computer-assisted diagnostic aid" (8)	1: Lab	347	Students	Non-contingent	Stated preferences
		2: Lab	128	Students	Non-contingent	Stated preferences
		3: Field	74	Hospital patients	Non-contingent	Stated preferences
		4: Lab	131	Medical students	Non-contingent	Stated preferences
Bahner et al., 2008	"Misuse of diagnostic aids in process control: the effects of automation misses on complacency and automation bias" (6, 9)	Lab	24	Students	Non-contingent	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
Bai et al., 2020	"The impacts of algorithmic work assignment on fairness perceptions and productivity: evidence from field experiments"	1: Field	50	Warehouse workers	Non-contingent	Stated, revealed preferences
		Replication: Field	20	Warehouse workers	Non-contingent	Stated, revealed preferences
Bartneck, Rosalia, et al., 2005	"Robot abuse – a limitation of the media equation"	Lab	20	Students & university employees	Non-contingent	Revealed preferences
Bartneck, Yogeeswaran, et al., 2018	"Robots and racism" (7)	1: Online	192	Crowdflower participants	contingent	Stated, revealed preferences
		2: Online	172	Crowdflower participants	contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Bickmore et al., 2005	"Establishing the computer – patient working alliance in automated health behavior change interventions" (8)	Online	91	Adults	Non-contingent	Stated, revealed preferences
Bigman and K. Gray, 2018	"People are averse to machines making moral decisions" (1, 2, 6, 10)	1: Online	242	MTurk participants	Non-contingent	Stated preferences
		2: Online	241	MTurk participants	Non-contingent	Stated preferences
		3: Online	240	MTurk participants	Non-contingent	Stated preferences
		4: Online	242	MTurk participants	Non-contingent	Stated preferences
		5: Online	485	MTurk participants	Non-contingent	Stated preferences
		6: Online	239	MTurk participants	Non-contingent	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
		7: Online	100	MTurk participants	Non-contingent	Revealed preferences
		8: Online	240	MTurk participants	Non-contingent	Stated preferences
		9 (within subjects): Online	201	MTurk participants	Non-contingent	Revealed preferences
		9 (between subjects): Online	482	MTurk participants	Non-contingent	Stated preferences
Briggs and Scheutz, 2014	"How robots can affect human behavior: Investigating the effects of robotic displays of protest and distress" (7)	1: Lab	20	students	N/A	Stated, revealed preferences
		2: Lab	13	students	N/A	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		3: Lab	14	students	N/A	Stated, revealed preferences
Bundorf et al., 2019	"How do humans interact with algorithms? Experimental evidence from health insurance"	RCT	1,185	Patients	Non-contingent	Stated, revealed preferences
Castelo, 2019	"Blurring the Line Between Human and Machine: Marketing Artificial Intelligence"	Ch. 3: 1: Online	387	MTurk participants	N/A	Stated preferences
		Ch. 3: 2: Online	41,592 views, 604 clicks	Facebook users	None	Revealed preferences
		Ch. 3: 3: Online	201	MTurk participants	N/A	Stated preferences
		Ch. 3: 4: Online	201	Prolific academic participants	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		Ch. 3: 5: Online	13,621 views, 101 clicks	Facebook users	None	Revealed preferences
		Ch. 3: 6: Online	399	Prolific academic participants	Contingent	Stated, revealed preferences
		Ch 4.: 1: Online	800	US Prolific academic participants	N/A	Stated preferences
		Ch 4.: 2: Online	100	MTurk participants	N/A	Stated, revealed preferences
		Ch 4.: 3: Online	282	Prolific academic participants	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		Ch 4.: 4: Online	300	Prolific academic participants	N/A	Stated preferences
		Ch 4.: 5: Lab	83	Students	N/A	Stated preferences & physiological measures
Chaminade et al., 2012	"How do we think machines think? an fMRI study of alleged competition with an artificial intelligence" (2, 8, 10)	Lab	19	Male students	Non-contingent	fMRI & stated preferences
Cohn et al., 2018	"Honesty in the digital age"	1: Online	486	Students	Contingent	Stated, revealed preferences
		2: Online	380	Students	Contingent	Stated, revealed preferences
Corgnet et al., 2019	"Rac(g)e Against the Machine?: Social Incentives When Humans Meet Robots"	Lab	240	University-educated young adults	Contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Coricelli and Nagel, 2009	"Neural correlates of depth of strategic reasoning in medial prefrontal cortex" (11)	Lab	20	Adults	Contingent	fMRI & stated preferences
Cormier et al., 2013	"Would you do as a robot commands? An obedience study for human-robot interaction" (7)	Lab	27	Adults	Non-contingent	Stated, revealed preferences
Correia et al., 2018	"Group-based emotions in teams of humans and robots" (7,9)	Lab	48	Students	Non-contingent	Stated preferences
Cowgill, 2018	"Bias and productivity in humans and algorithms: Theory and evidence from resume screening"	Field	N/A	HR professionals	N/A	Revealed preferences
Cowgill et al., 2020	"The managerial effects of algorithmic fairness activism"	1: Online	ca. 500	US adults	N/A	Stated preferences
		2: Online	ca. 500	US adults	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
De Laere et al., 1998	"The electronic mirror: human-computer interaction and change in self-appraisals" (1, 2, 7)	Lab	158	Students	Non-contingent	Stated preferences
Dell'Acqua et al., 2020	"Super Mario Meets AI: The Effects of Automation on Team Performance and Coordination in a Videogame Experiment"	Lab	220	N/A	Contingent	Stated, revealed preferences
Dietvorst and Bharti, 2019	"People reject algorithms in uncertain decision domains because they have diminishing sensitivity to forecasting error"	1: Online	601	MTurk participants	Contingent	Revealed preferences
		2: Online	403	MTurk participants	Contingent	Stated, revealed preferences
		3: Online	1,005	MTurk participants	Contingent	Stated, revealed preferences
		4: Online	405	MTurk participants	Contingent	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		5a: Online	401	MTurk participants	Contingent	Revealed preferences
		5b: Online	399	MTurk participants	Contingent	Revealed preferences
Dietvorst, Simmons, et al., 2015	"Algorithm aversion: people erroneously avoid algorithms after seeing them err." (2, 8, 10)	1: Lab	361	N/A	Contingent	Stated, revealed preferences
		2: Lab	206	N/A	Contingent	Stated, revealed preferences
		3a: Online	410	MTurk participants	Contingent	Stated, revealed preferences
		3b: Online	1,036	MTurk participants	Contingent	Stated, revealed preferences
		4: Lab	354	N/A	Contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Dietvorst, Simmons, et al., 2018	“Overcoming algorithm aversion: people will use imperfect algorithms if they can (even slightly) modify them” (4, 5)	1: Lab	288	N/A	Contingent	Stated, revealed preferences
		2: Online	816	MTurk participants	Contingent	Revealed preferences
		3: Online	818	MTurk participants	Contingent	Stated, revealed preferences
Dijkstra, 1999	“User agreement with incorrect expert system advice” (1, 2, 6, 7)	Lab	73	Students	Non-contingent	Stated preferences
Dijkstra et al., 1998	“Persuasiveness of expert systems” (1, 2, 6, 7)	Lab	85	Students	Non-contingent	Stated preferences
Domeinski et al., 2007	“Human redundancy in automation monitoring: effects of social loafing and social compensation” (6,9)	Lab	36	Students	N/A	Stated, revealed preferences
Dzindolet et al., 2002	“The perceived utility of human and automated aids in a visual detection task” (2, 6, 10)	1: Lab	68	Students	N/A	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
		2: Lab	128	Students	Contingent	Stated, revealed preferences
		3: Lab	71	Students	Contingent	Stated, revealed preferences
Efendić et al., 2020	“Slow response times undermine trust in algorithmic (but not human) predictions” (2, 4)	1:Online	304	MTurk participants	N/A	Stated preferences
		2: Online	302	MTurk participants	N/A	Stated preferences
		3: Online	486	MTurk participants	N/A	Stated preferences
		4a: Online	100	MTurk participants	N/A	Stated preferences
		4b: Online	100	MTurk participants	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		5: Online	236	Prolific participants	N/A	Stated preferences
		6: Online	200	MTurk participants	N/A	Stated preferences
		6, no info: Online	50	MTurk participants	N/A	Stated preferences
		7: Online	200	Prolific participants	Contingent	Stated, revealed preferences
Erlei et al., 2020	“Impact of Algorithmic Decision Making on Human Behavior: Evidence from Ultimatum Bargaining”	Online	1178	MTurk participants	Contingent	Stated, revealed preferences
Eyssel and Hegel, 2012	“(s)he’s got the look: Gender stereotyping of robots” (2)	Lab	60	Students	N/A	Stated, revealed preferences
Farjam and Kirchkamp, 2018	“Bubbles in hybrid markets: how expectations about algorithmic trading affect human trading” (3, 4)	Lab	216	Students	Contingent	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
Feier et al., 2021	"Hiding Behind Machines: When Blame Is Shifted to Artificial Agents"	Lab	149	Students	Contingent	Revealed preferences
Filiz et al., 2021	"Reducing algorithm aversion through experience" (3)	Lab	143	Students	Contingent	Revealed preferences
Fogg and C. Nass, 1997	"How users reciprocate to computers: an experiment that demonstrates behavior change" (7)	Lab	76	N/A	N/A	Revealed preferences
Galletta et al., 2005	"Does spell-checking software need a warning label?" (7)	Lab	65	Students	N/A	Revealed preferences
Galster et al., 2001	"Air traffic controller performance and workload under mature free flight: Conflict detection and resolution of aircraft self-separation" (2, 6, 7, 9)	Lab	10	Experts	None	Stated, revealed preferences
Goetz et al., 2003	"Matching robot appearance and behavior to tasks to improve human-robot cooperation" (7, 9)	1: Online	108	Students	N/A	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		2: Lab	21	Students	N/A	Stated, revealed preferences
		3: Online	47	Students	N/A	Stated, revealed preferences
Gogoll and Uhl, 2018	"Rage against the machine: automation in the moral domain" (2, 3, 6)	Lab	264	Students	Contingent	Stated, revealed preferences
Gombolay et al., 2015	"Decision-making authority, team efficiency and human worker satisfaction in mixed human–robot teams" (7)	1: Lab	N/A	Students and young professionals	N/A	Stated, revealed preferences
		2: Lab	N/A	Students and young professionals	N/A	Stated, revealed preferences
Gratch et al., 2007	"Creating rapport with virtual agents"	Lab	131	Adults	Non-contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
H. M. Gray et al., 2007	"Dimensions of mind perception" (1, 11)	Online	2,399	Adults interested in social and moral psychology	N/A	Revealed preferences
K. Gray and Wegner, 2012	"Feeling robots and human zombies: mind perception and the uncanny valley" (1, 2, 6, 10)	1: Lab-in-the-field	120	Adults, recruited in subway stations and campus dining halls	N/A	Stated preferences
		2: Lab-in-the-field	45	Adults, recruited in subway stations and campus dining halls	N/A	Stated preferences
		2b : Online	28	MTurk participants	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		3: Lab-in-the-field	44	Adults, recruited in subway stations and campus dining halls	N/A	Stated preferences
Hertz and Wiese, 2019	"Good advice is beyond all price, but what if it comes from a machine?" (2)	N/A	68	Students	Non-contingent	Revealed preferences
Hidalgo et al., 2021	*How humans judge machines*	Online	5903 in 89 studies	MTurk participants	N/A	Stated preferences
P. J. Hinds et al., 2004	"Whose job is it anyway? A study of human-robot interaction in a collaborative task" (2, 7)	Lab	292	Students	Non-contingent	Stated, revealed preferences
Hohenstein and M. Jung, 2020	"AI as a moral crumple zone: The effects of AI-mediated communication on attribution and trust" (1, 2, 7)	Online	113	Students	Non-contingent	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Humphreys et al., 2011	"Increasing discussions of intimate partner violence in prenatal care using Video Doctor plus Provider Cueing: a randomized, controlled trial" (6, 8, 12)	RCT	50	Pregnant women with IPV risk	N/A	Stated preferences
Ishowo-Oloko et al., 2019	"Behavioural evidence for a transparency–efficiency tradeoff in human–machine cooperation" (7)	Online	698	MTurk participants	Contingent	Revealed preferences
Jago, 2019	"Algorithms and authenticity" (4)	1: Online	175	Students	N/A	Stated preferences
		2: Online	401	MTurk participants	N/A	Stated preferences
		3: Online	200	MTurk participants	N/A	Stated preferences
		4: Online	804	MTurk participants	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
Kantowitz et al., 1997	"Driver acceptance of unreliable traffic information in familiar and unfamiliar settings" (2, 6, 10)	Lab	48	Young adults (able to drive)	Contingent	Stated, revealed preferences
Katagiri et al., 2001	"Cross-cultural studies of the computers are social actors paradigm: The case of reciprocity"	1: Lab	22+22	US and Japanese	N/A	Revealed preferences
		2: Lab	80	Japanese	N/A	Revealed preferences
Kim and P. Hinds, 2006	"Who should I blame? Effects of autonomy and transparency on attributions in human-robot interaction" (7)	Lab	157	Students	N/A	Stated preferences
Kirchkamp and Strobel, 2019	"Sharing responsibility with a machine" (2, 3, 6)	Lab	399	Students	Contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Kizilcec, 2016	“How much information? Effects of transparency on trust in an algorithmic interface” (7)	Field	103	Individuals enrolled in an open online course	Non-contingent	Stated preferences
Köbis and Mossink, 2021	“Artificial intelligence versus Maya Angelou: Experimental evidence that people cannot differentiate AI-generated from human-written poetry” (1, 2, 7)	1: Online	230	Prolific participants	Contingent	Stated, revealed preferences
		2: Online	600	Prolific participants	Contingent	Stated, revealed preferences
Krach et al., 2008	“Can machines think? interaction and perspective taking with robots investigated via fMRI” (11)	Lab	20	Male adults	Non-contingent	fMRI & stated, revealed preferences
E.-J. Lee, 2010	“What triggers social responses to flattering computers? Experimental tests of anthropomorphism and mindlessness explanations” (1, 6)	1: Lab	204	Students	Contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		2: Lab	149	Students	Contingent	Stated, revealed preferences
M. K. Lee, 2018	"Understanding perception of algorithmic decisions: fairness, trust, and emotion in response to algorithmic management" (5, 6, 7)	Online	228	MTurk participants (US residents)	Non-contingent	Stated preferences
M. K. Lee and Baykal, 2017	"Algorithmic mediation in group decisions: Fairness perceptions of algorithmically mediated vs. discussion-based social division" (7)	1: Lab	55	Participants recruited through a university-managed participant recruitment website	Non-contingent	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		2: Lab	103	Participants recruited through a university-managed participant recruitment website	Non-contingent	Stated preferences
M. K. Lee, Jain, et al., 2019	"Procedural justice in algorithmic fairness: leveraging transparency and outcome control for fair algorithmic mediation" (6, 7)	Lab	71	N/A	Non-contingent	Stated, revealed preferences
Leyer and Schneider, 2019	"Me, You or AI? How Do We Feel About Delegation" (7)	Online	1,246	Students, university employees and others	N/A	Stated, revealed preferences
Liu and Du, 2021	"Blame Attribution Asymmetry in Human–Automation Cooperation" (8, 9)	1a:N/A	240	Students	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
		1b:N/A	240	Students	N/A	Stated preferences
		2a: Lab	240	Students	N/A	Stated preferences
		2b: Online	325	Participants recruited through social media	N/A	Stated preferences
S. Lim and Reeves, 2010	"Computer agents versus avatars: responses to interactive game characters controlled by a computer or other player" (6, 7, 9)	Lab	34	Students	N/A	Stated preferences & physiological measures
Logg et al., 2019	"Algorithm appreciation: People prefer algorithmic to human judgment" (2, 4)	1a: Online	202	MTurk participants	Contingent	Revealed preferences
		1b: Online	215	MTurk participants	Contingent	Revealed preferences
		1c: Online	286	MTurk participants	N/A	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		1d: Online	119	Experts	Contingent	Revealed preferences
		2: N/A	154	N/A	N/A	Revealed preferences
		N/A	403	N/A	Contingent	Revealed preferences
		4: Online	70 experts, 301 MTurk partici-pants	Experts and MTurk participants	Contingent	Revealed preferences
Longoni et al., 2019	“Resistance to medical artificial intelligence” (1, 3, 4, 6)	1: N/A	228	Students	Non-contingent	Revealed preferences
		2: Online	103	MTurk participants	Non-contingent	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		3 (A-C): Online	744	MTurk participants	Non-contingent	Stated preferences
		4: Online	100	MTurk participants	Non-contingent	Revealed preferences
		5: Online	286	MTurk participants	Non-contingent	Stated preferences
		6: Online	243	MTurk participants	Non-contingent	Stated preferences
		7: Online	294	MTurk participants	Non-contingent	Stated preferences
		8: Online	401	MTurk participants	Non-contingent	Stated preferences
		9: Online	197	MTurk participants	Non-contingent	Stated preferences
Lucas et al., 2014	"It's only a computer: virtual humans increase willingness to disclose" (1, 2, 7)	Lab	154	Adults	N/A	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
Luo et al., 2019	"Frontiers: machines vs. humans: The impact of artificial intelligence chatbot disclosure on customer purchases" (3, 4)	Field	6200	Customers of a financial service company	None	Revealed preferences
Mandryk et al., 2006	"Using psychophysiological techniques to measure user experience with entertainment technologies" (1, 2, 6, 7)	1: Lab	7	Male students	N/A	Stated preferences & physiological measures
		2: Lab	10	Students	N/A	Stated preferences & physiological measures
Manzey et al., 2012	"Human performance consequences of automated decision aids: the impact of degree of automation and system experience" (2, 6, 7, 9)	1: Lab	56	Students	Non-contingent	Stated, revealed preferences
		2: Lab	88	Students	Non-contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
McCabe et al., 2001	"A functional imaging study of cooperation in two-person reciprocal exchange" (11)	Lab	12	N/A	Contingent	fMRI
Melo, Carnevale, et al., 2011	"The effect of expression of anger and happiness in computer agents on negotiations with humans" (7,9)	Lab	150	Students	Contingent	Stated, revealed preferences
Melo, Marsella, et al., 2016	"People do not feel guilty about exploiting machines" (7)	1: Online	81	MTurk participants	Contingent	Stated, revealed preferences
		2: N/A	165	Students	Contingent	Revealed preferences
Mende et al., 2019	"Service robots rising: how humanoid robots influence service experiences and elicit compensatory consumer responses" (3, 4)	1a: Lab	80	Students	Non-contingent	Revealed preferences
		1b: Lab	253	Students	Non-contingent	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
		1c: Lab	215	Students	Non-contingent	Revealed preferences
		2: Online	100	MTurk participants	N/A	Stated, revealed preferences
		3a: Online	180	MTurk participants	N/A	Revealed preferences
		3b: Lab	203	Students	Non-contingent	Revealed preferences
		4: Lab	250	Students	Non-contingent	Revealed preferences
Metzger and Parasuraman, 2005	"Automation in future air traffic management: effects of decision aid reliability on controller performance and mental workload" (2, 6, 10)	1: Lab	12	Experts	Non-contingent	Stated, revealed preferences
		2: Lab	12	Experts	Non-contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Metzger, Duley, et al., 2000	"Effects of variable-priority training on automation-related complacency: performance and eye movements" (6, 9)	Lab	43	Students	Non-contingent	Revealed preferences & eye movement data
Mirnig et al., 2017	"To err is robot: How humans assess and act toward an erroneous social robot" (7)	Lab	45	Adults	N/A	Stated, revealed preferences
Moon, 2000	"Intimate exchanges: using computers to elicit self-disclosure from consumers" (1,3,4,6)	1: Lab	60	Students	Non-contingent	Stated, revealed preferences
		2: Lab	24	Students	Non-contingent	Stated, revealed preferences
Moon, 2003	"Don't blame the computer: when self-disclosure moderates the self-serving bias" (2, 4)	1: Lab	48	Adults	Non-contingent	Stated, revealed preferences
		2: Lab	62	Adults	Non-contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Moon and C. Nass, 1998	"Are computers scapegoats? Attributions of responsibility in human – computer interaction" (6, 7, 9)	Lab	80	Students	Non-contingent	Stated preferences
Mosier, Palmer, et al., 1992	"Electronic checklists: implications for decision making" (6, 9)	Lab	24	Experts	N/A	Stated, revealed preferences
Mosier, Skitka, Heers, et al., 1998	"Automation bias: decision making and performance in high-tech cockpits" (2, 6, 7, 9)	Lab	25	Experts	N/A	Stated, revealed preferences
Mosier, Skitka, Dunbar, et al., 2001	"Aircrews and automation bias: the advantages of teamwork?" (2, 6, 7, 9)	Lab	48	Commercial pilots	N/A	Stated, revealed preferences
Muir and Moray, 1996	"Trust in automation. Part II. Experimental studies of trust and human intervention in a process control simulation" (6, 13)	1: Lab	6	Students (Male)	Contingent	Stated, revealed preferences
		2: Lab	6	Students (Male)	Contingent	Stated, revealed preferences

Table 1: Experimental studies

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Naquin and Kurtzberg, 2004	"Human reactions to technological failure: How accidents rooted in technology vs. human error influence judgments of organizational accountability" (2,4)	1: Lab	86	Students	N/A	Stated preferences
		2: Lab	89	Students	N/A	Stated preferences
C. Nass, Fogg, et al., 1996	"Can computers be teammates?" (6, 7, 9)	Lab	56	Students	N/A	Stated, revealed preferences
C. Nass, Green, et al., 1997	"Are machines gender neutral? Gender-stereotypic responses to computers with voices" (2)	Lab	40	Students	N/A	Stated preferences
Newman et al., 2020	"When eliminating bias isn't fair: Algorithmic reductionism and procedural justice in human resource decisions" (2, 4)	1: Online	199	MTurk participants	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
		2: Field	1654	Employees from a large private university	Non-contingent	Stated preferences
		3: Online	189	MTurk participants	N/A	Stated preferences
		4: N/A	197	Students	N/A	Stated preferences
		5: Lab	213	Students	Non-contingent	Stated preferences
Nishio et al., 2018	“Do robot appearance and speech affect people’s attitude? Evaluation through the ultimatum game”	Lab	21	Students	Contingent	Revealed preferences
Nitsch and Glassen, 2015	“Investigating the effects of robot behavior and attitude towards technology on social human-robot interactions” (2,6,7)	Lab	48	Opportunity sample	Contingent	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Önkal et al., 2009	“The relative influence of advice from human experts and statistical methods on forecast adjustments” (1, 2, 4, 5, 6)	1: Lab	76	Students	Non-contingent	Stated, revealed preferences
		2: Lab	54	Students	Non-contingent	Stated, revealed preferences
Palmeira and Spassova, 2015	“Consumer reactions to professionals who use decision aids” (4)	1: Online	70	US Adults	Non-contingent	Stated preferences
		2: Online	192	US Adults	Non-contingent	Stated preferences
		3: Online	83	US Adults	Non-contingent	Stated preferences
Parasuraman, Molloy, et al., 1993	“Performance consequences of automation-induced ’complacency’” (2, 6, 7, 9)	1: Lab	24	Adults	Non-contingent	Revealed preferences
		2: Lab	16	Adults	Non-contingent	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Paravisini and Schoar, 2013	"The incentive effect of scores: randomized evidence from credit committees"	RCT	1421	Credit committee members of a bank	N/A	Revealed preferences
M. V. Pezzo and S. P. Pezzo, 2006	"Physician evaluation after medical errors: does having a computer decision aid help or hurt in hindsight?" (8)	1: N/A	59	Students	Non-contingent	Stated preferences
		2: N/A	320	Medical and non-medical students	Non-contingent	Stated preferences
Prahl and Van Swol, 2017	"Understanding algorithm aversion: when is advice from automation discounted?" (4, 5, 7, 14)	Online	157	Students	Non-contingent	Stated, revealed preferences
Promberger and Baron, 2006	"Do patients trust computers?" (1, 2, 4, 5, 6)	1: Online	86	Adults	Non-contingent	Stated preferences
		2: Online	80	Adults	Non-contingent	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Ribeiro et al., 2016	“” Why should i trust you?” Explaining the predictions of any classifier” (7)	1: Online	400	MTurk participants (with basic knowledge about religion)	N/A	Revealed preferences
		2: Online	10	MTurk participants	N/A	Stated, revealed preferences
Rosenthal-von der Pütten et al., 2013	“An experimental study on emotional reactions towards a robot” (7)	Lab	18	Students	Non-contingent	Stated, revealed preferences
Rovira et al., 2007	“Effects of imperfect automation on decision making in a simulated command and control task” (2, 6, 10)	Lab	18	Students	Non-contingent	Stated, revealed preferences
Salem et al., 2015	“Would you trust a (faulty) robot? Effects of error, task type and personality on human-robot cooperation and trust” (7,9)	Lab	40	Adults	N/A	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Salomons et al., 2018	"Humans conform to robots: Disambiguating trust, truth, and conformity" (7,9)	Lab	30	Students	Non-contingent	Stated, revealed preferences
Sanfey et al., 2003	"The neural basis of economic decision – making in the ultimatum game" (1, 11)	Lab	19	N/A	Contingent	Revealed preferences & fMRI
Sarter and Schroeder, 2001	"Supporting decision making and action selection under time pressure and uncertainty: the case of in-flight icing" (2, 6, 10)	Lab	27	Commercial pilots	Non-contingent	Revealed preferences
Saygin et al., 2012	"The thing that should not be: predictive coding and the uncanny valley in perceiving human and humanoid robot actions" (2, 8, 10)	Lab	20	Adults	N/A	fMRI
Schniter et al., 2020	"Trust in humans and robots: economically similar but emotionally different" (2, 3, 6)	Lab	387	Students	Contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Shaffer et al., 2013	"Why do patients derogate physicians who use a computer-based diagnostic support system?" (8)	1: N/A	434	Students	Non-contingent	Stated preferences
		2: Online	109	Students	Non-contingent	Stated preferences
		3: N/A	189	Students	Non-contingent	Stated preferences
Shank, 2012	"Perceived justice and reactions to coercive computers" (6)	Lab	114	Students	Contingent	Stated, revealed preferences
Shank et al., 2019	"When are artificial intelligence versus human agents faulted for wrongdoing? Moral attributions after individual and joint decisions" (6)	Online	453	MTurk participants	Non-contingent	Stated preferences
Short, Hart, et al., 2010	"No fair!! an interaction with a cheating robot" (7)	Lab	60	Students	N/A	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Short and Mataric, 2017	"Robot moderation of a collaborative game: Towards socially assistive robotics in group interactions" (7)	Lab	30	Freshman IT students	N/A	Stated, revealed preferences
Skitka, Mosier, and Burdick, 1999	"Does automation bias decision-making?" (6, 7, 9)	Lab	80	Students	Non-contingent	Stated, revealed preferences
Skitka, Mosier, and Burdick, 2000	"Accountability and automation bias" (6, 7, 9)	Lab	181	Students	Non-contingent	Stated, revealed preferences
Skitka, Mosier, Burdick, and Rosenblatt, 2000	"Automation bias and errors: are crews better than individuals?" (2, 6, 7, 9)	Lab	144	Students	Non-contingent	Revealed preferences
Slater et al., 2006	"A virtual reprise of the Stanley Milgram obedience experiments" (11)	Lab	34	Students & University Employees	N/A	Stated, revealed preferences & physiological measures
Sproull et al., 1996	"When the interface is a face" (2,7)	Lab	130	Students	Non-contingent	Stated, revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	**Title and Discipline(s)**	**Type**	**N**	**Subjects**	**Incentives**	**Elicitation measures**
Strobel, 2019	"The hidden costs of automation"	Online	520	MTurk participants	Contingent	Revealed preferences
Strohkorb et al., 2016	"Improving human-human collaboration between children with a social robot" (2, 6, 7)	Lab	86	Children (6-9 y.o.)	N/A	Stated, revealed preferences
Tay, Park, et al., 2013	"When stereotypes meet robots: the effect of gender stereotypes on people's acceptance of a security robot"	Lab	40	Students	Non-contingent	Stated preferences
Tay, Y. Jung, et al., 2014	"When stereotypes meet robots: the double-edge sword of robot gender and personality in human–robot interaction" (1, 2, 7)	Lab	164	Students	N/A	Stated, revealed preferences
Tan et al., 2018	"Investigating human+ machine complementarity for recidivism predictions"	Online	20	MTurk participants	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Tazelaar and Snijders, 2004	“The myth of purchasing professionals’ expertise. More evidence on whether computers can make better procurement decisions” (4)	Original: N/A	91	Students & purchasing professionals	N/A	Stated preferences
		I&L test - 1: online	72	Mostly purchasing managers	N/A	Stated preferences
		I&L test - 2: online	72	Mostly purchasing managers	N/A	Stated preferences
		The Mas-terclass: lab	13	Purchasing professionals	N/A	Stated preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		Replication and extension: N/A	118	Students & purchasing professionals & people with no knowledge about purchasing	N/A	Stated preferences
Teubner et al., 2015	"The impact of computerized agents on immediate emotions, overall arousal and bidding behavior in electronic auctions" (7)	Lab	120	Students	Contingent	Revealed preferences & physiological measures
Traeger et al., 2020	"Vulnerable robots positively shape human conversational dynamics in a human–robot team" (11)	Lab	153	N/A	non-contingent	revealed and stated
Tsai et al., 2003	"Computer decision support as a source of interpretation error: the case of electrocardiograms" (8)	RCT	30	Physicians	N/A	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
Wout et al., 2006	“Affective state and decision-making in the ultimatum game” (10)	Lab	30	Students	Contingent	Revealed preferences & physiological measures
Von der Puetten et al., 2010	““It doesn’t matter what you are!” explaining social effects of agents and avatars.” (1, 2, 7)	Lab	83	Adults	Non-contingent	Revealed preferences
Waytz and Norton, 2014	“Botsourcing and outsourcing: robot, British, Chinese, and German workers are for thinking — not feeling — jobs.” (2, 8)	1: Online	103	MTurk participants	Non-contingent	Stated preferences
		2: Online	266	MTurk participants	Non-contingent	Stated preferences
		3: Online	54	MTurk participants	Non-contingent	Stated preferences
		4: Online	153	MTurk participants	Non-contingent	Stated preferences

Table 1: Experimental studies

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		5: Online	167	MTurk participants	Non-contingent	Stated preferences
		6a: Online	166	MTurk participants	Non-contingent	Stated preferences
		6b: Online	167	MTurk participants	Non-contingent	Stated preferences
		6c: Online	164	MTurk participants	Non-contingent	Stated preferences
Waytz, Heafner, et al., 2014	"The mind in the machine: anthropomorphism increases trust in an autonomous vehicle" (2, 6)	Lab	100	Adults (able to drive)	N/A	Stated preferences & physiological measures
Wickens et al., 2015	"Complacency and automation bias in the use of imperfect automation" (2, 6, 10)	Lab	119	Students	Non-contingent	Revealed preferences
Yeomans et al., 2019	"Making sense of recommendations" (1, 2, 4, 5, 6)	1a: N/A	122	Museum visitors	N/A	Revealed preferences

Table 1: EXPERIMENTAL STUDIES

Author	Title and Discipline(s)	Type	N	Subjects	Incentives	Elicitation measures
		1b: Online	544	MTurk participants	Contingent	Revealed preferences
		2: N/A	210	Museum visitors	N/A	Revealed preferences
		3: Online	886	MTurk participants	N/A	Stated, revealed preferences
		4: Online	899	MTurk participants	N/A	Stated, revealed preferences
		5: Online	972	MTurk participants	N/A	Stated, revealed preferences
Zhang et al., 2010	“Service robot feature design effects on user perceptions and emotional responses” (7, 9)	Lab in the field	24	Seniors (64-91 y.o.)	N/A	Stated, revealed measures

Table 2: OBSERVATIONAL STUDIES

Author	Title	Data	Topic
Broek, 2019	"Hiring algorithms: an ethnography of fairness in practice"	Large European FMCG company	Use of an AI tool in hiring decisions
Gates et al., 2002	"Automated underwriting in mortgage lending: good news for the underserved?"	Federal Home Loan Mortgage Corporation	Use of automated underwriting systems in mortgage lending decisions
Hoffman et al., 2018	"Discretion in hiring"	Personnel data of 15 firms in service sector with low-skilled workers	Identification of human biases in hiring by comparison of actual hiring decisions with automated recommendations
Kleinberg et al., 2018	"Human decisions and machine predictions"	Arrests in New York City	Use of machine learning in judicial recidivity risk decisions
Lebovitz et al., 2019	"Doubting the diagnosis: how artificial intelligence increases ambiguity during professional decision making"	Department of radiology in a US hospital	Use of AI tools in diagnostic decision-making in radiology
Saez de Tejada Cuenca, 2019	"Essays on Social and Behavioral Aspects of Apparel Supply Chains"	Large clothing retailer	Incorporation of decision aid support regarding price-setting in sales decisions by managers
Stevenson, 2018	"Assessing risk assessment in action"	Criminal cases in Kentucky	Use of an algorithmic risk assessment tool in the criminal justice system
Stubbs et al., 2007	"Autonomy and common ground in human-robot interaction: a field study"	Observations of scientific field work (Life in the Atacama project)	Use of robot to support scientific research

Table 3: LITERATURE REVIEWS

Author	Title	Topic
Alberdi, Strigini, et al., 2009	"Why are people's decisions sometimes worse with computer support?"	Automation bias and mechanisms involving human errors when using computer support

Table 3: LITERATURE REVIEWS

Author	Title	Topic
Burton et al., 2020	"A systematic review of algorithm aversion in augmented decision making"	Algorithm aversion with the focus on conditions that lead to the acceptance or rejection of algorithmically generated insights
Glikson and Woolley, 2020	"Human trust in Artificial Intelligence: Review of empirical research"	The determinants of human trust in AI
Goddard et al., 2012	"Automation bias: a systematic review of frequency, effect mediators, and mitigators"	Automation bias with focus on frequency, effect mediators, and mitigators
Jussupow et al., 2020	"Why are we averse towards Algorithms? A comprehensive literature Review on Algorithm aversion."	Algorithm aversion and algorithm appreciation
Köbis, Bonnefon, et al., 2021	"Bad machines corrupt good morals"	How AI influences ethical behaviour of humans.
J. D. Lee and See, 2004	"Trust in automation: designing for appropriate reliance"	Link between trust and reliance on automation
V. Lim et al., 2021	"Social robots on a global stage: establishing a role for culture during human–robot interaction"	The role of cultural background on reactions and expectations towards social robots.
March, 2019	"The behavioral economics of artificial intelligence: Lessons from experiments with computer players"	Review of experimental studies that use computer players
Mende et al., 2019	"Service robots rising: how humanoid robots influence service experiences and elicit compensatory consumer responses"	Use of robots in service settings (part of the paper)
Mosier and Skitka, 1996	"Human decision makers and automated decision aids: made for each other?"	Human interactions with automated decision aids
C. Nass and Moon, 2000	"Machines and mindlessness: social responses to computers"	Summary of a series of experimental studies that demonstrate that individuals mindlessly apply social rules and expectations to computers
Parasuraman and Riley, 1997	"Humans and automation: use, misuse, disuse, abuse"	Theoretical, empirical, and analytical studies pertaining to human use, misuse, disuse, and abuse of automation technology

Table 3: LITERATURE REVIEWS

Author	**Title**	**Topic**
Parasuraman and Manzey, 2010	“Complacency and bias in human use of automation: an attentional integration”	Complacency and bias in interactions with automated decision support systems
Złotowski et al., 2015	“Anthropomorphism: opportunities and challenges in human–robot interaction”	Summary of potential benefits and challenges of building anthropomorphic robots, from both a philosophical perspective and from the viewpoint of empirical research in the fields of human–robot interaction and social psychology

B Key findings and related papers of each section

Table 4: OVERVIEW AND KEY FINDINGS OF SUBSECTION 2.1: THE PERCEPTION OF AUTOMATED AGENTS AS SOCIAL INTERACTION PARTNERS

Key findings	Key related studies
Human-AA interactions as social interactions	De Laere et al., 1998; Fogg and C. Nass, 1997; Katagiri et al., 2001; C. Nass, Fogg, et al., 1996; C. Nass, Green, et al., 1997; C. Nass and Moon, 2000; Reeves and C. I. Nass, 1996; Aharoni and Fridlund, 2007; Bartneck, Yogeeswaran, et al., 2018; Eyssel and Hegel, 2012; E.-J. Lee, 2010; Salomons et al., 2018; Tay, Y. Jung, et al., 2014; Tay, Park, et al., 2013; Von der Puetten et al., 2010
Different areas of brain activated in Human-AA interactions, less brain activity to infer the mental state of AA	Chaminade et al., 2012; Coricelli and Nagel, 2009; Krach et al., 2008; McCabe et al., 2001; Sanfey et al., 2003
No attribution of 'mind' w.r.t. AA	H. M. Gray et al., 2007; K. Gray and Wegner, 2012
But aversion to physical or psychological mistreatment of AA	Bartneck, Rosalia, et al., 2005; Briggs and Scheutz, 2014; Rosenthal-von der Pütten et al., 2013; Slater et al., 2006

Table 5: OVERVIEW AND KEY FINDINGS OF SUBSECTION 2.2: THE REDUCED EMOTIONAL AND SOCIAL RESPONSE TO AUTOMATED AGENTS

Key findings	Key related studies
Reduced emotional and social response in human-AA interactions	Adam, Krämer, et al., 2015; S. Lim and Reeves, 2010; Mandryk et al., 2006; Melo, Marsella, et al., 2016; Sanfey et al., 2003; Schniter et al., 2020; Teubner et al., 2015; Wout et al., 2006
Narrowed emotional spectrum; less positive and less negative reactions in human-AA interactions	Leyer and Schneider, 2019; M. V. Pezzo and S. P. Pezzo, 2006; Shank, 2012

Potential cause: lower/no perception of intentionality of AA	Hidalgo et al., 2021; Shank, 2012
Increased economic rationality in human-AA interactions	Adam, Teubner, et al., 2018; Erlei et al., 2020; Farjam and Kirchkamp, 2018
Increased willingness to disclose sensitive information to AA	Ahmad et al., 2009; Humphreys et al., 2011; Lucas et al., 2014
Less pro-social reactions towards AA, more self-serving or unethical behavior,	Cohn et al., 2018; Corgnet et al., 2019; Köbis, Bonnefon, et al., 2021; Melo, Marsella, et al., 2016; Moon, 2003; Moon and C. Nass, 1998
The intensity of emotional and social responce depends on the use (e.g., culture) and be amplified by appearance and behavior of AA	Bickmore et al., 2005; Castelo, 2019; Correia et al., 2018; Gratch et al., 2007; K. Gray and Wegner, 2012; P. J. Hinds et al., 2004; E.-J. Lee, 2010; V. Lim et al., 2021; Mende et al., 2019; Nishio et al., 2018; Nitsch and Glassen, 2015; Short, Hart, et al., 2010; Zhang et al., 2010

Table 6: OVERVIEW AND KEY FINDINGS OF SUBSECTION 2.3: THE IMPORTANCE OF TASK TYPE

Key findings	Key related studies
Replacing humans with AA in (some) interactions can cause strong negative reactions	Ishowo-Oloko et al., 2019; Luo et al., 2019
Preference for humans over AA in moral and social tasks, but less/none in analytical tasks	Castelo, 2019; Hertz and Wiese, 2019; Jago, 2019; Köbis and Mossink, 2021; M. K. Lee, 2018; Waytz and Norton, 2014; cf. subsection 3.2 and 4.4

Table 7: OVERVIEW AND KEY FINDINGS OF SUBSECTION 3.1: AUTOMATED AGENTS AS PEERS

Key findings	Key related studies
Willingness to collaborate with AA depends on capabilities of AA and on cognitive abilities of humans	Alekseev, 2020; Dell'Acqua et al., 2020
Negative performance effects due to reduced social response and increased coordination failures	Corgnet et al., 2019; Dell'Acqua et al., 2020
Positive performance effects due to higher perceived responsibility of a human for a task	Gombolay et al., 2015; Paravisini and Schoar, 2013
Little or no deflection of blame by human to automated teammates	Hohenstein and M. Jung, 2020; Kirchkamp and Strobel, 2019
Third party observers fault AA rather than a human teammate for bad outcomes	Feier et al., 2021; Liu and Du, 2021; Naquin and Kurtzberg, 2004; M. V. Pezzo and S. P. Pezzo, 2006; Shank et al., 2019
Level of autonomy of automated agents affects attribution of blame in hybrid teams	Kim and P. Hinds, 2006
Presence of automated teammates affects interactions among humans in team	Short and Mataric, 2017; Strohkorb et al., 2016; Traeger et al., 2020

Table 8: OVERVIEW AND KEY FINDINGS OF SUBSECTION 3.2: AUTOMATED AGENTS AS SUPERIORS

Key findings	Key related studies
Willingness to accept and follow automated managers	Bai et al., 2020; Cormier et al., 2013; Gombolay et al., 2015
Acceptance of automated managers depends on the type of the task (analytical vs. social)	M. K. Lee, 2018; Newman et al., 2020

Table 9: OVERVIEW AND KEY FINDINGS OF SUBSECTION 4.1: THE AVERSION TO AUTOMATED AGENTS AS DELEGATES

Key findings	Key related studies
Algorithm aversion as preference to rely on own judgment rather than delegating to (superior) AA	Dietvorst, Simmons, et al., 2015; Dzindolet et al., 2002; Kantowitz et al., 1997
Algorithm aversion as preference for delegating to human rather than AA	Hertz and Wiese, 2019; Longoni et al., 2019; Promberger and Baron, 2006; Yeomans et al., 2019
Low trust in AA as potential reason for algorithm aversion	Dietvorst and Bharti, 2019; Longoni et al., 2019; Önkal et al., 2009; Parasuraman and Riley, 1997; Saez de Tejada Cuenca, 2019
More leniency to human mistakes as compared to algorithmic ones as potential reason for algorithm aversion	Dietvorst and Bharti, 2019; Dietvorst, Simmons, et al., 2015, 2018; Dzindolet et al., 2002; Muir and Moray, 1996; Prahl and Van Swol, 2017
Algorithm aversion occurring irrespective of perceived quality of AA	Gogoll and Uhl, 2018; Longoni et al., 2019; Mirnig et al., 2017; Salem et al., 2015
Algorithm aversion in moral tasks/ in tasks perceived to require individual deliberation	Bigman and K. Gray, 2018; Gogoll and Uhl, 2018; Longoni et al., 2019
Algorithm aversion in analytical tasks	Dietvorst and Bharti, 2019; Dietvorst, Simmons, et al., 2015, 2018; Önkal et al., 2009
Indifference between decision-making support by human or AA; preference for AA over human support	Prahl and Van Swol, 2017, generally cf. sections 4.2 and 4.3

Table 10: OVERVIEW AND KEY FINDINGS OF SUBSECTION 4.2: THE OVER-RELIANCE ON AUTOMATED DECISION AIDS

Key findings	Key related studies
Automation bias as over-reliance on inputs by automated decision-making support	Alberdi, Povyakalo, Strigini, and Ayton, 2004; Alberdi, Povyakalo, Strigini, Ayton, and Given-Wilson, 2008; Bahner et al., 2008; Galletta et al., 2005; Galster et al., 2001; Manzey et al., 2012; Mosier, Palmer, et al., 1992; Mosier and Skitka, 1996; Mosier, Skitka, Heers, et al., 1998; Rovira et al., 2007; Sarter and Schroeder, 2001; Skitka, Mosier, and Burdick, 1999; Tsai et al., 2003; Wickens et al., 2015, reviews: Alberdi, Strigini, et al., 2009; Parasuraman and Manzey, 2010
Automation bias may occur irrespective of level of expertise of the user	Galster et al., 2001; Metzger and Parasuraman, 2005; Mosier, Skitka, Heers, et al., 1998; Sarter and Schroeder, 2001
Higher likelihood of automation bias in situations with cognitive load	Mosier, Skitka, Dunbar, et al., 2001; Parasuraman, Molloy, et al., 1993; Skitka, Mosier, Burdick, and Rosenblatt, 2000
Means to curb automation bias: Attention training, highlighting responsibility and accountability	Goddard et al., 2012; Metzger, Duley, et al., 2000; Skitka, Mosier, and Burdick, 2000

Table 11: OVERVIEW AND KEY FINDINGS OF SUBSECTION 4.3: THE PREFERENCE FOR AUTOMATED DECISION AIDS

Key findings	Key related studies
Algorithm appreciation as preference for receiving and following automated rather than human advice	Dijkstra, 1999; Dijkstra et al., 1998; Logg et al., 2019

Less/no algorithm aversion or algorithm appreciation if able to intervene and adjust the recommendation	Bigman and K. Gray, 2018; Dietvorst, Simmons, et al., 2018; Longoni et al., 2019; Palmeira and Spassova, 2015; Promberger and Baron, 2006